IATAI 2017

Illinois Association of Technical Accident Investigators

IATAI

Crash Data Review

CRASH DATA SPECIALISTS
WWW.CDR-TRAINERS.COM

ACCIDENT ANALYSIS & RECONSTRUCTION, INC

crash™
Collision Reconstruction And Scientific Hypothesis LLC

Crash Test Crew

- Brad Muir - Crash Data Specialists
- Jon Northrup- Crash Data Specialists
- Vlad Bortchevsky – Crash LLC
- Greg Russell – Accident Analysis & Reconstruction
- Anthony Barnes – North Carolina Highway Patrol

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ACCIDENT ANALYSIS
& RECONSTRUCTION, INC

TROOPER
NORTH CAROLINA STATE HIGHWAY PATROL

Instrumentation

- IST – Instrumented Sensor Technology
  - EDR3C / 3D Tri-axial accelerometers sampling at 3072hz
  - EDR3D Motion Master Tri-axial accelerometer at 3072hz
    - Includes 3 – axis angular rate



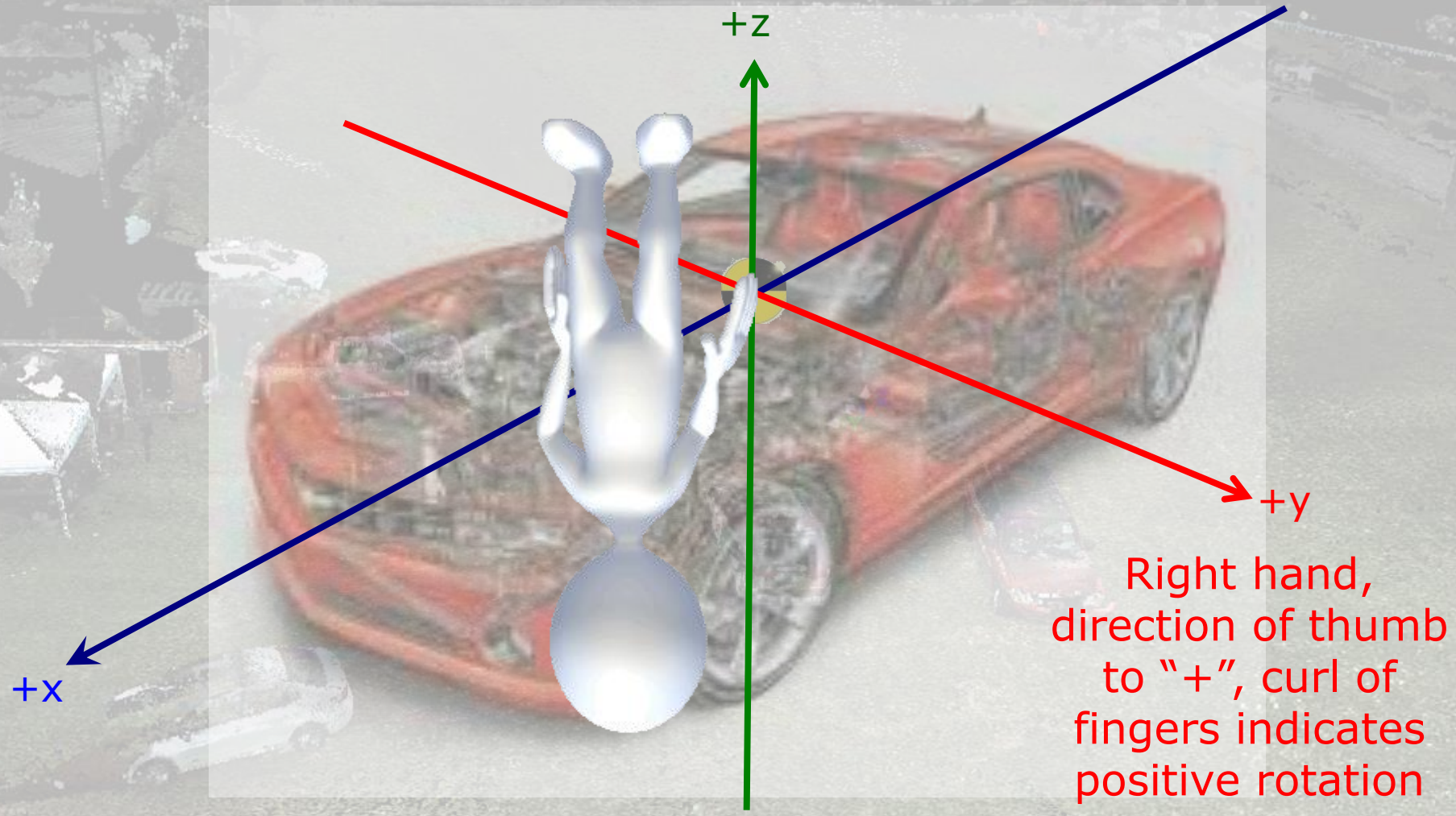


Instrumentation

- Garmin Virb XE
  - At least 3 per vehicle
  - 1 on each side
  - 120 FPS
  - LED Brake Light for sync
  - GPS Speed recorded



IST – Polarity (ISO)



Right hand, direction of thumb to "+", curl of fingers indicates positive rotation

The Plan!!!

- Crash Test series on actual city streets (not parking lots!)
- Construct a series of crash tests to explore relevant Reconstruction concepts:
  - Momentum
  - Closing Speed
  - Using Delta-v and PDOF
- Examine Relevant CDR data based on collision setup and dynamics and compare to independent instrumentation and traditional Reconstruction methods

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Active Safety Systems

- What may appear to be anomalous data may be activation of these systems, i.e.
  - Automatic Braking
  - Lane Centering





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Pre-Collision Braking System

(g) Emergency Brake Signal (w/ Emergency Brake Signal)

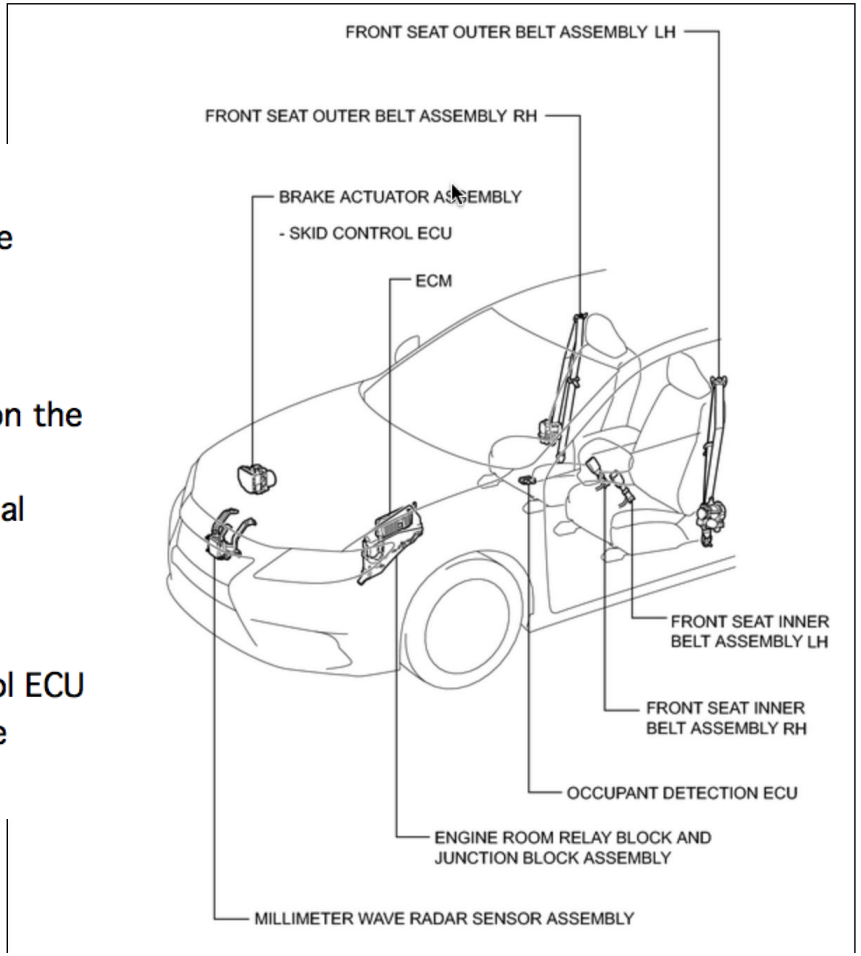
(1) In the case of emergency braking, the emergency brake signal flashes the stop lights to alert the drivers in following vehicles. The stop lights flash at approximately 4.0 Hz.

(h) Dynamic Radar Cruise Control (w/ Dynamic Radar Cruise Control)

(1) When the driving support ECU assembly determines that a large deceleration is required based on the distance to the preceding vehicle and the relative vehicle speed during the dynamic radar cruise control system operation, the skid control ECU operates the brake system according to the signal from the driving support ECU assembly.

(i) Pre-collision Control (w/ Pre-collision Control)

(1) When the driving support ECU assembly determines a high possibility of collision, the skid control ECU activates the pre-collision brake assist or the pre-collision brake according to the signal from the driving support ECU assembly.



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Pre-Crash Data

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 1)

Time (sec)	-4.55	-4.05	-3.55	-3.05	-2.55	-2.05	-1.55	-1.05	-0.55	-0.05	0 (TRG)
Vehicle Speed (MPH [km/h])	19.9 [32]	19.9 [32]	19.9 [32]	20.5 [33]	20.5 [33]	18.6 [30]	9.3 [15]	5 [8]	0.6 [1]	0 [0]	0 [0]
Accelerator Pedal, % Full (%)	13.5	18.0	23.5	21.5	14.5	23.5	67.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	3.0	4.0	4.5	5.5	3.0	4.5	0.0	3.0	0.0	0.0	0.0
Engine RPM (RPM)	1,300	1,300	1,300	1,300	1,300	1,200	1,000	1,800	900	700	700
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.44	4.37	7.87	6.34
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.359	0.287	0.287	0.287	0.502	-4.450	-8.973	-1.005	-4.307	8.973	8.973
YawRate (deg/sec)	0.00	-0.49	0.00	0.00	-0.49	-0.49	1.46	-0.49	-0.49	0.00	3.90
Steering Input (degrees)	3	3	3	3	3	3	3	6	15	24	27
Shift Position	D	D	D	D	D	D	D	D	D	D	D

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2016 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2016 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Optional on Limited and Limited Hybrid models with Toyota Safety Sense™ P option (TY)	3554 – 4Dr. Sedan Limited V6 3514 – 4Dr. Sedan Limited Hybrid
2016 RAV4 and RAV4 Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard on Limited and Limited Hybrid. Optional on XLE and XLE Hybrid models with Convenience Package (CP) and on SE with Advanced Technology Package (AT)	4450 – 5Dr. Wagon Limited 4452 – 5Dr. Wagon Limited AWD 4454 – 5Dr. Wagon Limited Hybrid 4440 – 5Dr. Wagon XLE 4442 – 5Dr. Wagon XLE AWD 4444 – 5Dr. Wagon XLE Hybrid 4470 – 5Dr. Wagon SE 4472 – 5Dr. Wagon SE AWD
2016 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard on Prius Three Touring and Prius Four Touring. Optional on Prius Three and Prius Four with Advanced Technology Package (AT)	1226 – 5Dr. Sedan Three Touring 1228 – 5Dr. Sedan Four Touring 1225 – 5Dr. Sedan Three 1227 – 5Dr. Sedan Four
2016 Prius *c*	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Optional on Prius c Three and Prius c Four with any additional package: Prius c Three with Deluxe Moonroof and Safety Package or Wheel and Safety Package. Prius c Four with Upgraded Wheel and Safety Package	1205 – 5Dr. Hatchback Three 1207 – 5Dr. Hatchback Four

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VEHICLE	TSS-C or TSS-P	FITMENT / AVAILABILITY	MODEL NUMBERS
2017 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2017 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Corolla		Standard	All Model Numbers
2017 Highlander and Highlander Hybrid		Standard	All Model Numbers
2017 RAV4 and RAV4 Hybrid		Standard	All Model Numbers
2017 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Prius Prime		Standard	All Model Numbers
2017 Corolla iM	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Standard	All Model Numbers
2017 Prius *c*		Standard	All Model Numbers
2017 Yaris		Standard	All Model Numbers

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PRE-COLLISION SYSTEM WITH PEDESTRIAN DETECTION FUNCTION (PCS W/PD) – VEHICLE AND PEDESTRIAN DETECTION

As there is a limit to the degree of recognition accuracy and control performance that this system can provide, do not overly rely on this system. This system will not prevent collisions or lessen collision damage or injury in every situation. Do not use PCS[3] instead of normal braking operations under any circumstances. Do not attempt to test the operation of the pre-collision system[3] yourself, as the system may not operate or engage, possibly leading to an accident. In some situations such as when driving in inclement weather (heavy rain, fog, snow,sandstorm, etc.) or while driving on a curve and for a few seconds after driving on a curve, a preceding vehicle/pedestrian may not be detected by the radar and camera sensors, preventing the system from operating properly.



VEHICLE DETECTION

With Toyota Safety Sense™ P[2], PCS[3] uses an **in-vehicle camera and front-grill mounted millimeter-wave radar** to help detect the vehicle in front of your vehicle.

- When the PCS[3] determines that the possibility of a frontal collision with that vehicle is high, it prompts the driver to take evasive action and brake, by using an audio and visual alert
  - **These alerts operate when the vehicle speed is between approximately 7 to 110 MPH for potential collisions with a vehicle**
- If the driver notices the hazard and brakes, the system may provide additional braking force using Brake Assist[4]. This system may apply greater braking force in relation to how strongly the brake pedal is depressed
- If the driver does not brake in a set time and the system determines that the possibility of a frontal collision with another vehicle is extremely high, the system may automatically apply the brakes, reducing speed in order to help the driver reduce the impact and in certain cases avoid the collision
  - **The PCS[3] included with the TSS-P[2] package may operate automated braking for potential collisions with a vehicle when vehicle speeds are between approximately 7 to 110 MPH[11]**
  - **May reduce vehicle speed by up to 25 MPH for potential collisions with a vehicle[11]**

CDR STATUS NO Interface

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Vehicle Control History

- Available on:
- 2013+ Rav 4 / 2014 IS250 / 350
- More added in 2015 +
- Requires Toyota TechStream Diagnostic Tool
  - Available with the “Lite” Version









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Vehicle Control History

- Specific Triggers result in CASA (ACM) Storing Pre / Post Trigger Data
- 26 Different “Behaviours” will result in Trigger
- 3 Categories:
  - Vehicle Input - Accelerator Pedal Input
  - Control System Activation (VCS / ABS etc..)
  - Vehicle Motion (Long. / Lat. g load)

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Vehicle Control History

- Up to 140 VCH Events can be stored
  - Stored in “FIFO” method – first in, first out
  - Recording memory is “partitioned”
    - Prevents rare triggers from being overwritten by more frequent ones
- 5-8 seconds of Pre & Post trigger data
  - Trigger Type determines time frame
- Up to 40 parameters including Odometer
- If equipped with Nav System, date & time

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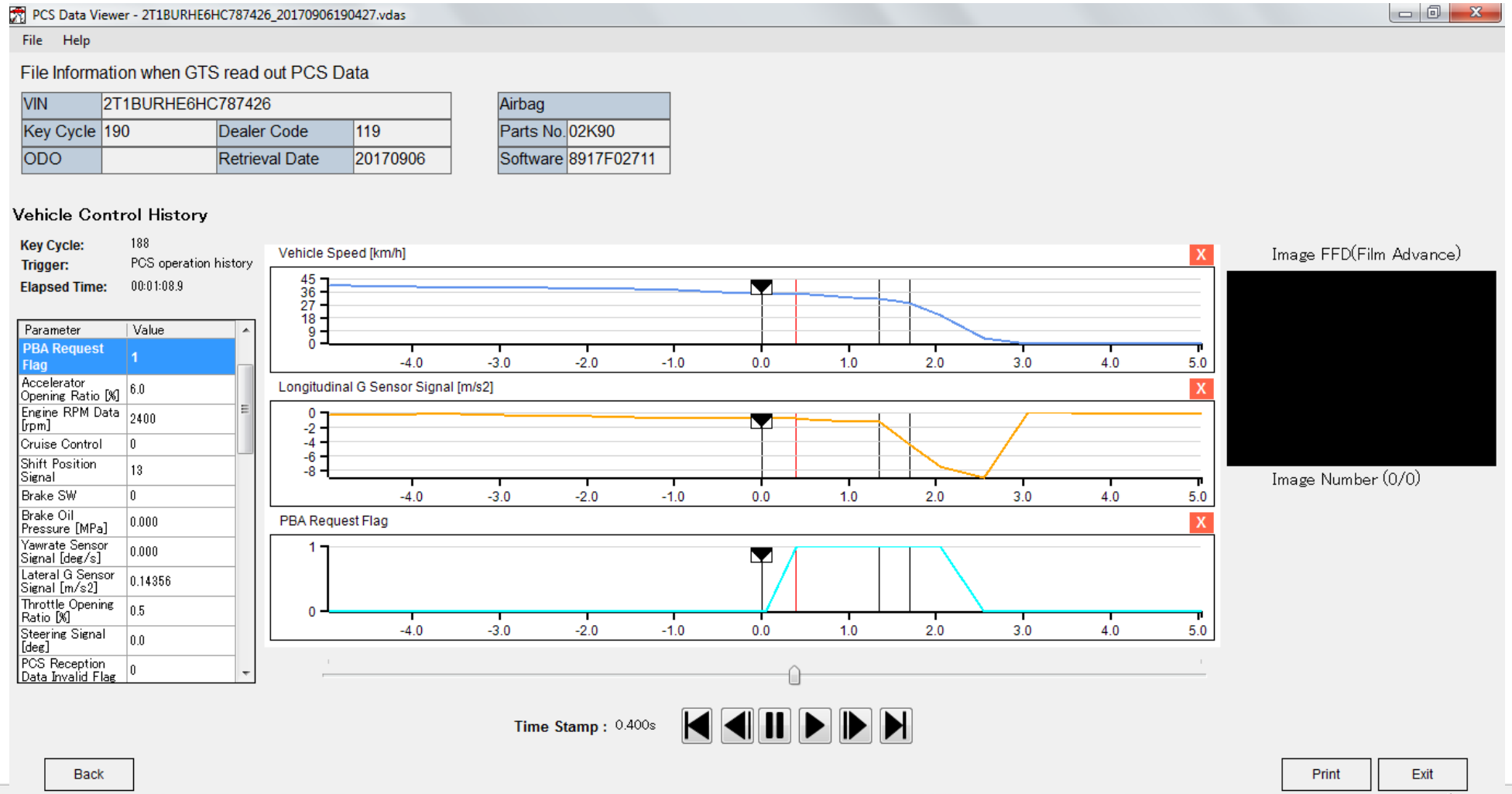


CDR STATUS

NO Interface

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PCS Data Viewer



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GM Automatic Low Speed Braking In Action





IATAI 2017 - Crash Test #1

2002 GMC Envoy vs. 2006 Buick Rendezvous

2006 Buick Rendezvous

Actual Weight: 3829lbs – 4N6XPRT AutoStats Weight: 4024lbs



2006 Buick Rendezvous

Actual Weight: 3829lbs – 4N6XPRT AutoStats Weight: 4024lbs



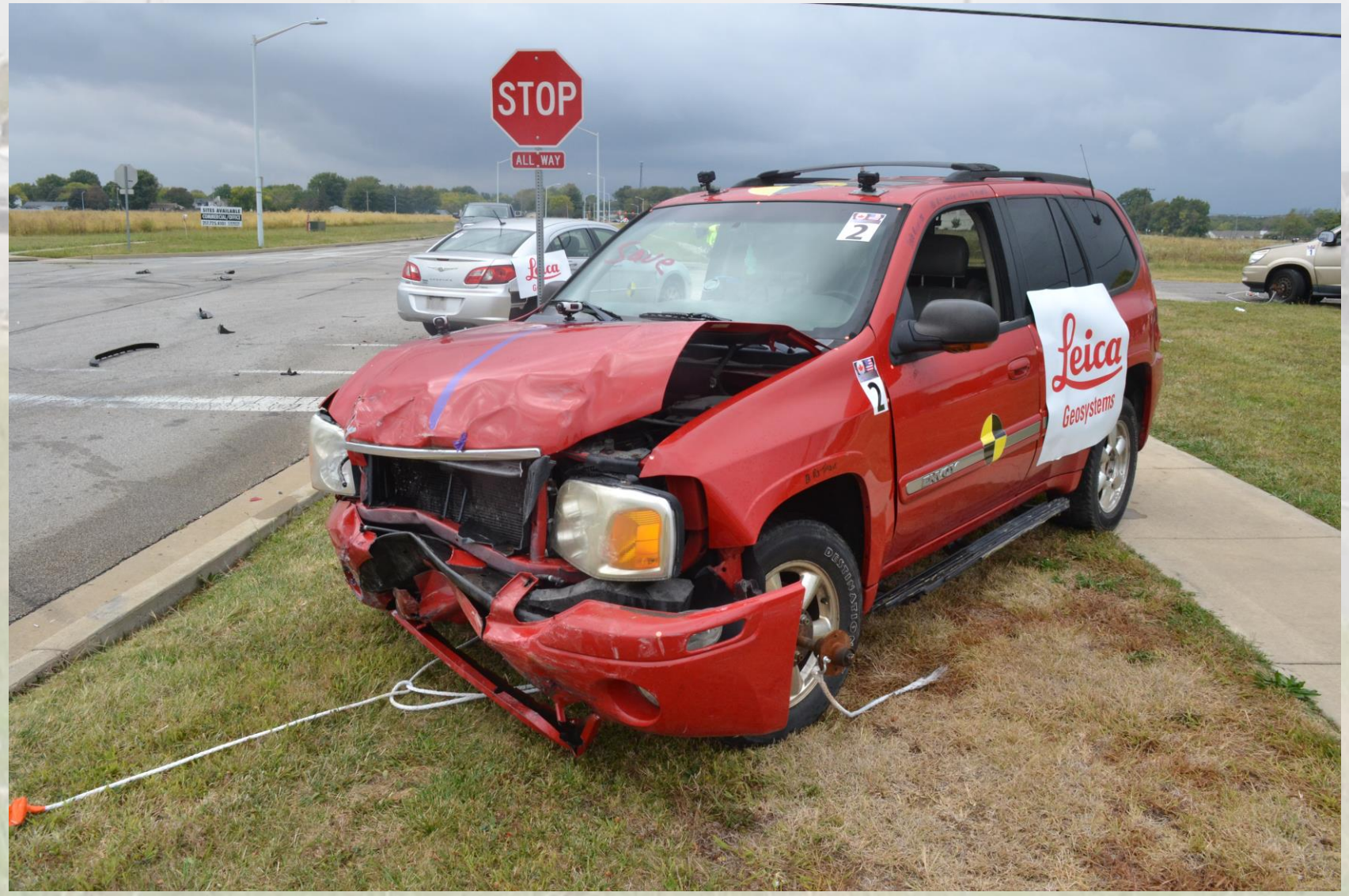
2002 GMC Envoy

Actual Weight: 4586lbs – 4N6XPRT AutoStats Weight: 4417lbs



2002 GMC Envoy

Actual Weight: 4586lbs – 4N6XPRT AutoStats Weight: 4417lbs



The Scene…….



Pix4Dmapper Pro - iatai-crash-1a

Project | Process | View | Navigation | Clipping | Point Cloud Editing

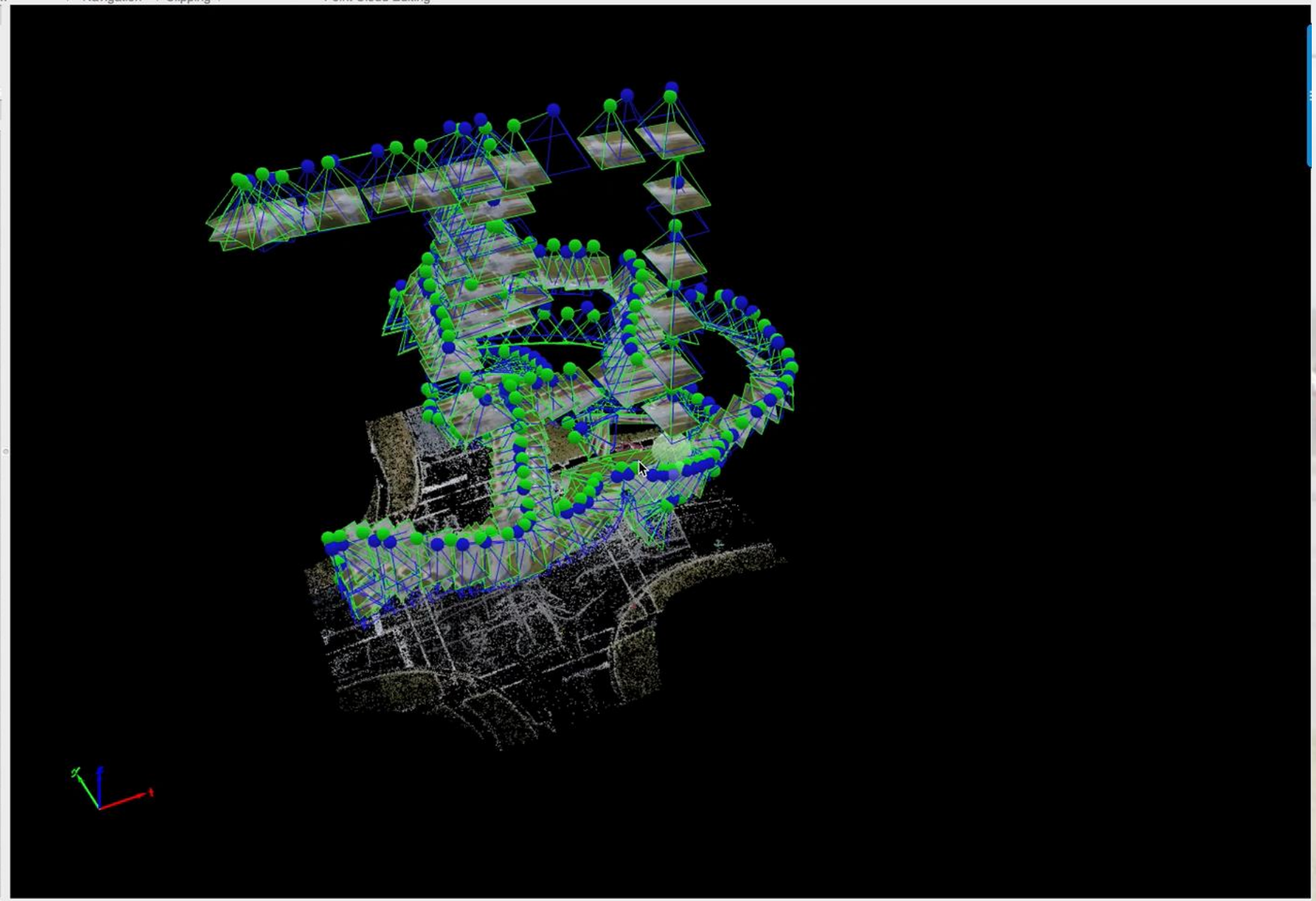
Unclassified | Assign

Home
Map View
rayCloud
Volumes
Mosaic Editor
Index Calculator
Processing
Log Output
Processing Options

Create

Layers

- Cameras
- Rays
- Tie Points
  - GCPs / MTPs
  - Automatic
- Processing Area
- Point Clouds
- Point Groups
- Triangle Meshes
- Objects



NAD_1983_StatePlane_Illinois_West_FIPS_1202_Feet (egm96) - (2452552.36, 1130226.48, 260.39) [ft]

Pix4Dmapper Pro - iatai-crash-1a

Project | Process | View | Navigation | Clipping | Point Cloud Editing

Unclassified | Assign

Home
Map View
rayCloud
Volumes
Mosaic Editor
Index Calculator
Processing
Log Output
Processing Options

Create

Layers

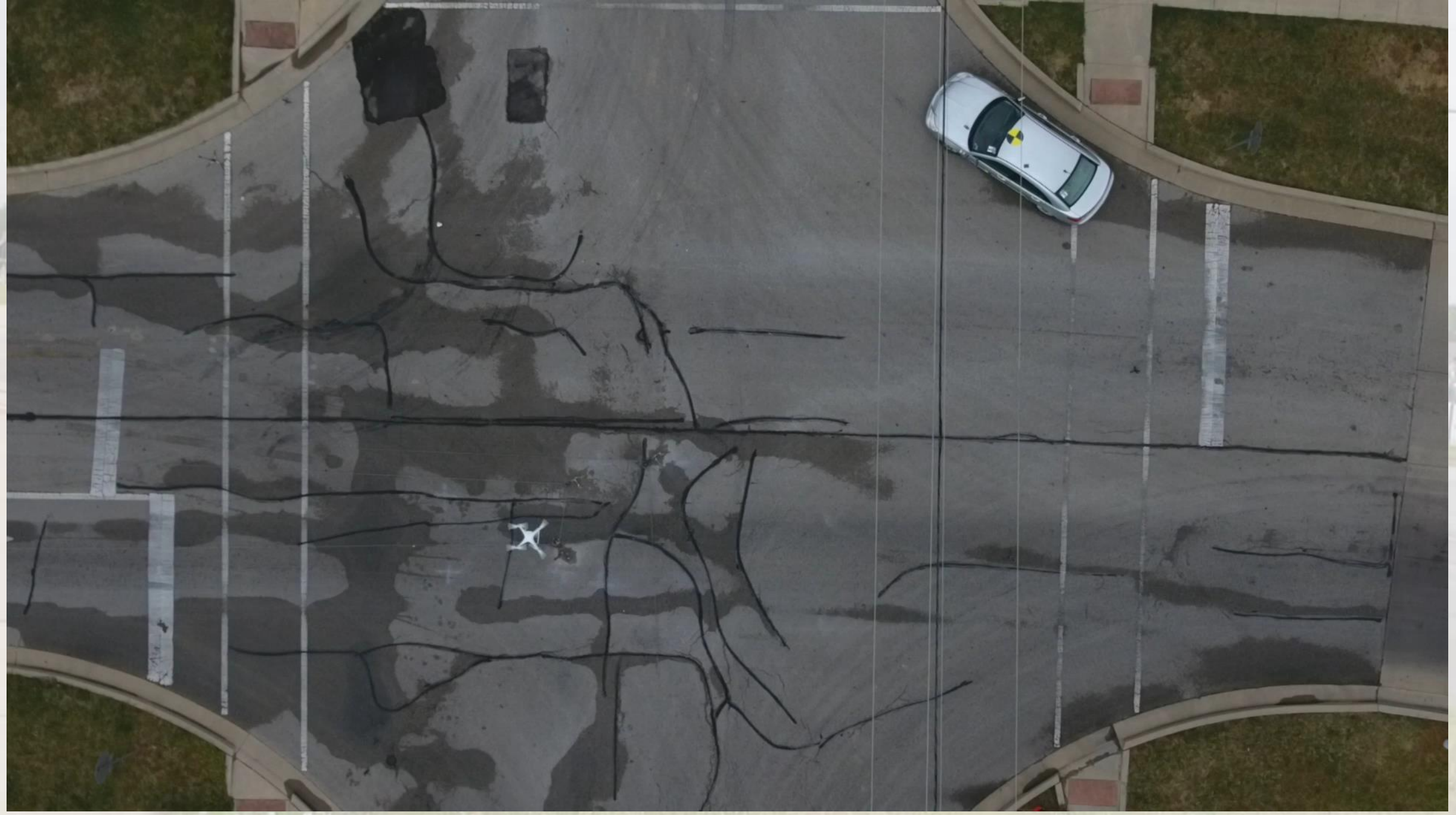
- Cameras
- Rays
- Tie Points
  - GCPs / MTPs
  - Automatic
- Processing Area
- Point Clouds
  - Densified Point Cloud
    - Display Properties
      - Point Size
      - Shader: Screen Aligned Qua
    - iatai-crash-1a_group1_c
- Point Groups
  - Unclassified
  - Deleted
- Triangle Meshes
- Objects



Layer Mesh iatai-crash-1a_simplified_3d_mesh loaded

NAD_1983_StatePlane_Illinois_West_FIPS_1202_Feet (egm96) - (2452531.36, 1130223.52, 224.93) [ft]


STOP



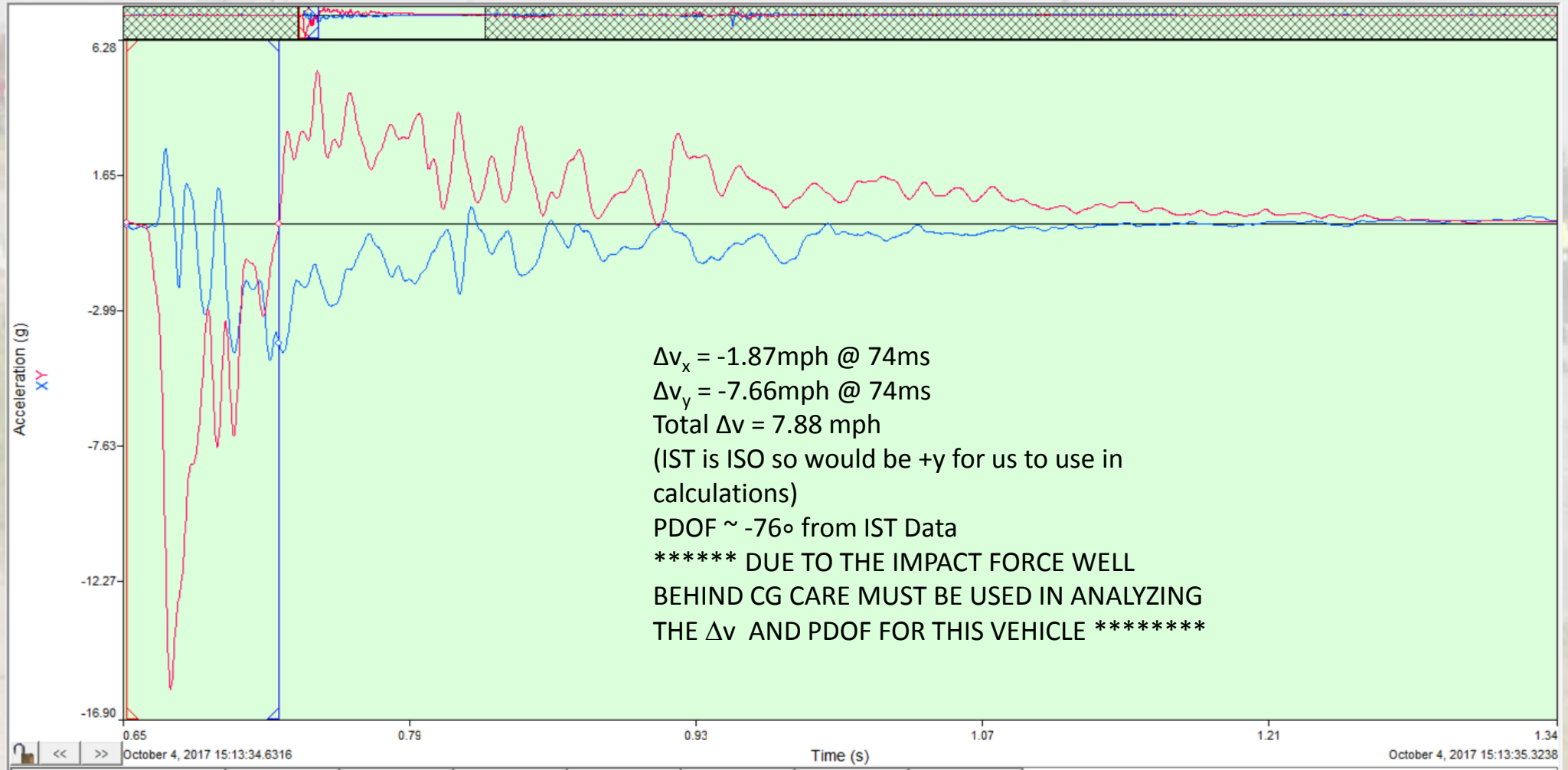





0
36
72
108
144
180
215
251
287
323
359
395
431
5
mph

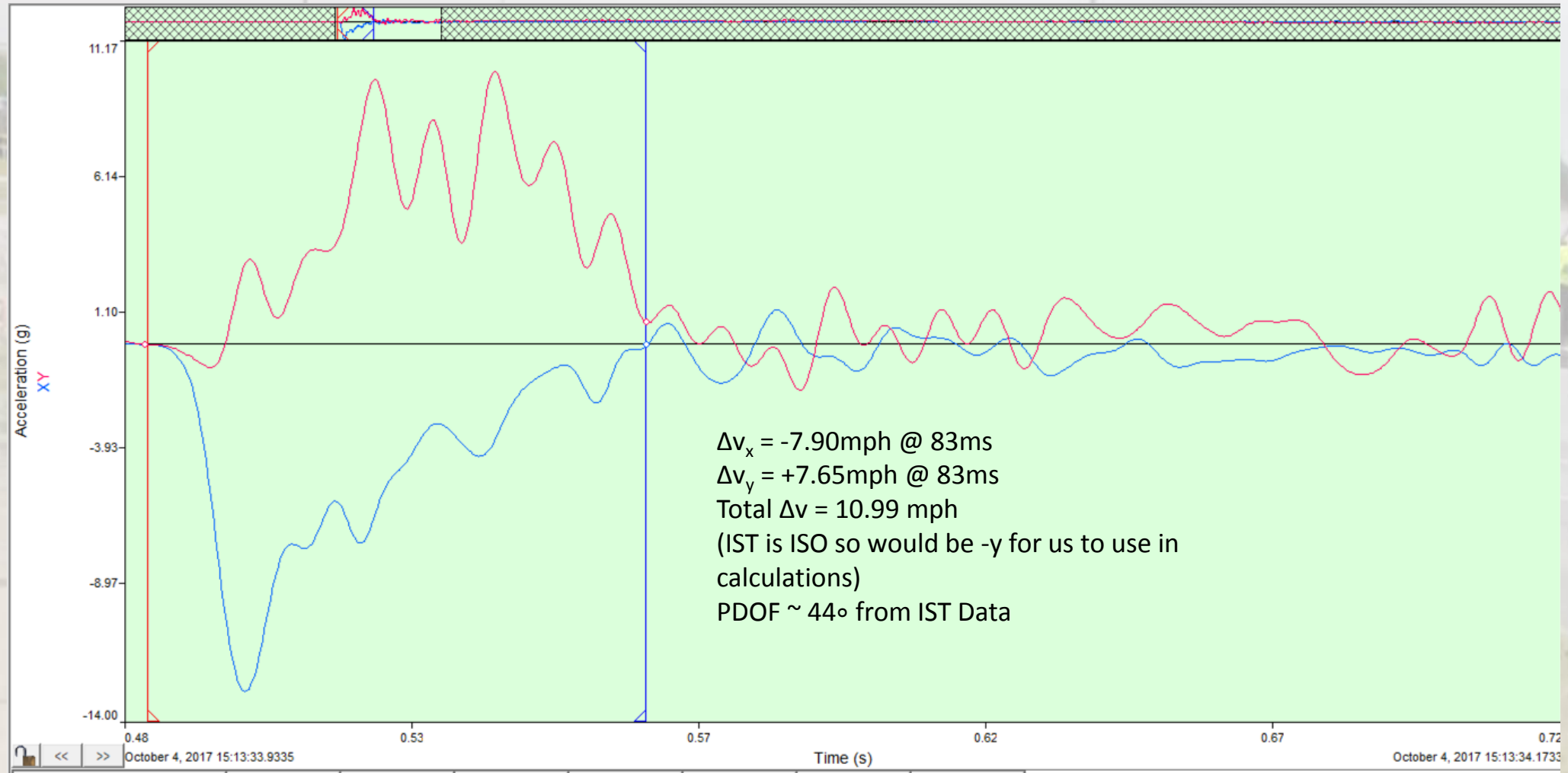

231
193
270
154
308
116
347
77
385
39
424
0
462
0
MPH
215
179
250
143
286
107
322
72
358
36
394
0
429
2
mph

IST - Rendezvous



Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f EDR3D #3045A 1:1:X	2013	-0.04944	2238	-4.044	-1.869	0.07324	13.65
f EDR3D #3045A 1:1:Y	2013	0.04664	2238	0.02038	-7.656	0.07324	13.65

IST - Envoy



Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f EDR3C #2211A 1:1:X	1478	-0.05085	1735	-0.06456	-7.899	0.08366	11.95
f EDR3C #2211A 1:1:Y	1478	-0.04757	1735	0.7561	7.648	0.08366	11.95

CDR Data - Rendezvous

CDR File Information

User Entered VIN	3G5DA03L06S622319
User	B. Muir CDS LLC
Case Number	IATAI 2017 - Post-Crash #1
EDR Data Imaging Date	10/04/2017
Crash Date	10/04/2017
Filename	2006 BUICK RENDEVOUS - IATAI 2017 - POST-CRASH TEST #1 - 3G5DA03L06S622319_ACM.CDRX
Saved on	Wednesday, October 4 2017 at 18:29:30
Imaged with CDR version	Crash Data Retrieval Tool 17.4.2
Imaged with Software Licensed to (Company Name)	Crash Data Specialists
Reported with CDR version	Crash Data Retrieval Tool 17.4.2
Reported with Software Licensed to (Company Name)	Crash Data Specialists
EDR Device Type	Airbag Control Module
Event(s) recovered	Non-Deployment

01014_SDMDW_r004

CDR Data - Rendezvous

- For THIS ACM type – the next occurring event ***may*** overwrite an existing non-deployment
- No mention of severity

Data Limitations

Recorded Crash Events:

There are two types of Recorded Crash Events. The first is the Non-Deployment Event. A Non-Deployment Event records data but does not deploy the air bag(s). It contains Pre-Crash and Crash data. The SDM can store up to one Non-Deployment Event. This event may be overwritten by another Non-Deployment Event. This event will be cleared by the SDM, after approximately 250 ignition cycles. This event can be overwritten by a second Deployment Event, referred to as a Deployment Level Event, if the Non-Deployment Event is not locked. The data in the Non-Deployment Event file will be locked, if the Non-Deployment Event occurred within five seconds before a Deployment Event. A locked Non Deployment Event cannot be overwritten or cleared by the SDM.

CDR Data - Rendezvous

- Passenger Seatbelt Status WAS buckled for this crash

System Status At Non-Deployment

SIR Warning Lamp Status	OFF
Driver's Belt Switch Circuit Status	BUCKLED
Passenger Belt Switch Circuit Status (If Equipped)	BUCKLED
Driver Seat Position Status (If Equipped)	Rearward
Passenger Seat Position Status (If Equipped)	Rearward
Ignition Cycles At Non-Deployment	38079
Ignition Cycles At Investigation	38081
Maximum SDM Recorded Velocity Change (MPH)	0.00
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	62.5
Time Between this Event and the Previous Event (sec)	0.1
Event Recording Complete	Yes

Time Between indicates multiple Non-Deployment events observed

CDR Data - Rendezvous

- Pre-existing Non-Deployment – “UNBUCKLED”
- When viewed with the Deployment from the crash test, for this vehicle, the “If Equipped” means it *WAS* monitored

System Status At Non-Deployment

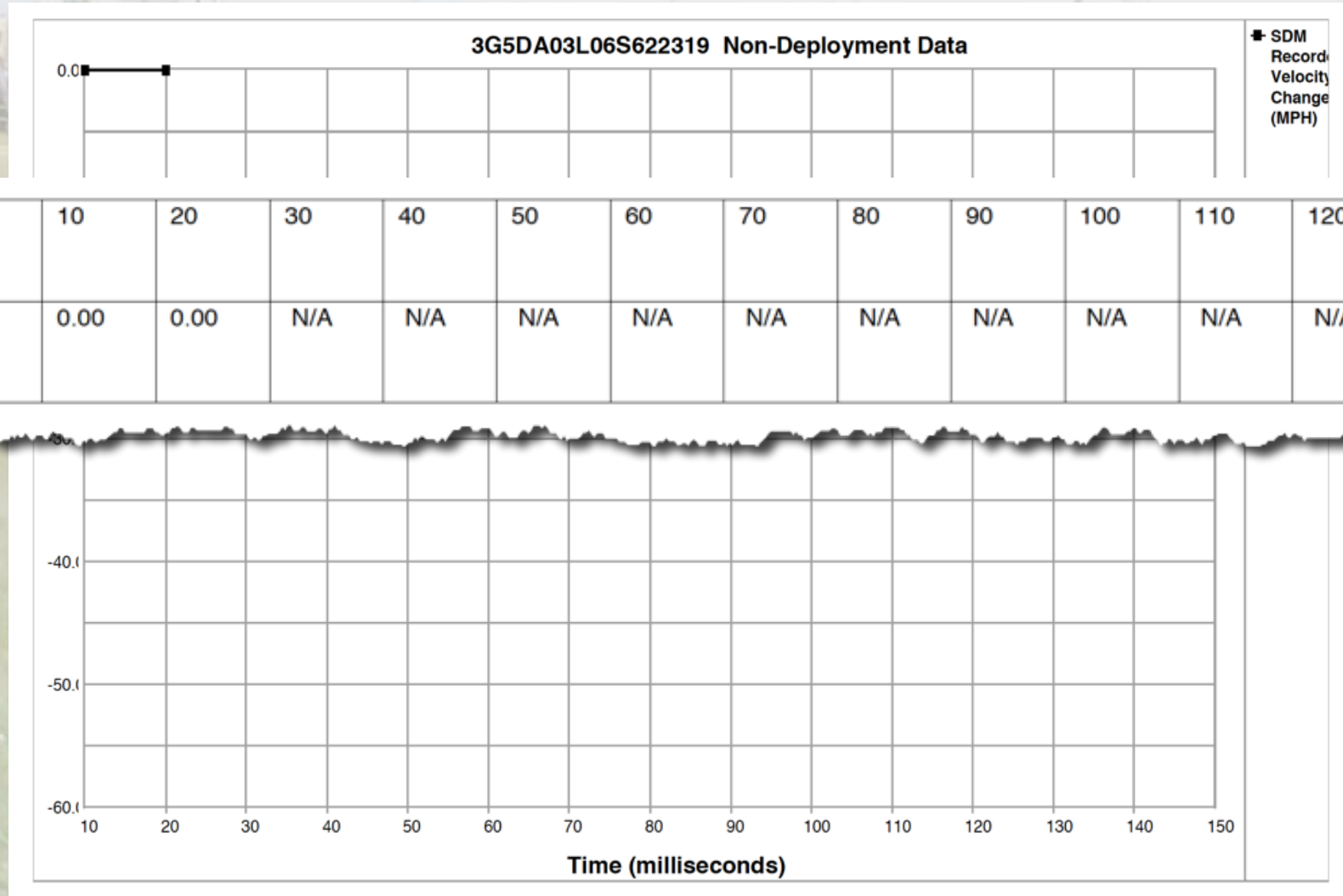
SIR Warning Lamp Status	OFF
Driver's Belt Switch Circuit Status	UNBUCKLED
Passenger Belt Switch Circuit Status (If Equipped)	UNBUCKLED
Driver Seat Position Status (If Equipped)	Rearward
Passenger Seat Position Status (If Equipped)	Rearward
Ignition Cycles At Non-Deployment	38029
Ignition Cycles At Investigation	38071
Maximum SDM Recorded Velocity Change (MPH)	-0.32
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	27.5
Time Between this Event and the Previous Event (sec)	N/A
Event Recording Complete	Yes

CDR Data - Rendezvous

- We know the impact speed was ~38 mph
- Coupled with the “Time between this event” data we can conclude multiple Non-Deployment events were observed and THIS event was triggered during the post impact travel from the main impact

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle	Brake Switch Circuit State
				ON
				ON
				ON
-5	32	1280	0	ON
-4	35	1344	0	ON
-3	38	1344	0	ON
-2	17	1408	0	ON
-1	2	896	0	ON

CDR Data - Rendezvous



Time (milliseconds)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
SDM Recorded Velocity Change	0.00	0.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

CDR Data - Envoy

CDR File Information

User Entered VIN	1GKDT13S822326698
User	B. Muir CDS LLC
Case Number	IATAI 2017
EDR Data Imaging Date	10/04/2017
Crash Date	10/04/2017
Filename	2002 GMC ENVOY - IATAI 2017 - POST-CRASH #1-1GKDT13S822326698_ACM.CDRX
Saved on	Wednesday, October 4 2017 at 18:34:39
Imaged with CDR version	Crash Data Retrieval Tool 17.4.2
Imaged with Software Licensed to (Company Name)	Crash Data Specialists
Reported with CDR version	Crash Data Retrieval Tool 17.4.2
Reported with Software Licensed to (Company Name)	Crash Data Specialists
EDR Device Type	Airbag Control Module
Event(s) recovered	Deployment

01030_SDMGT-2002_r005

CDR Data - Envoy

System Status At Deployment

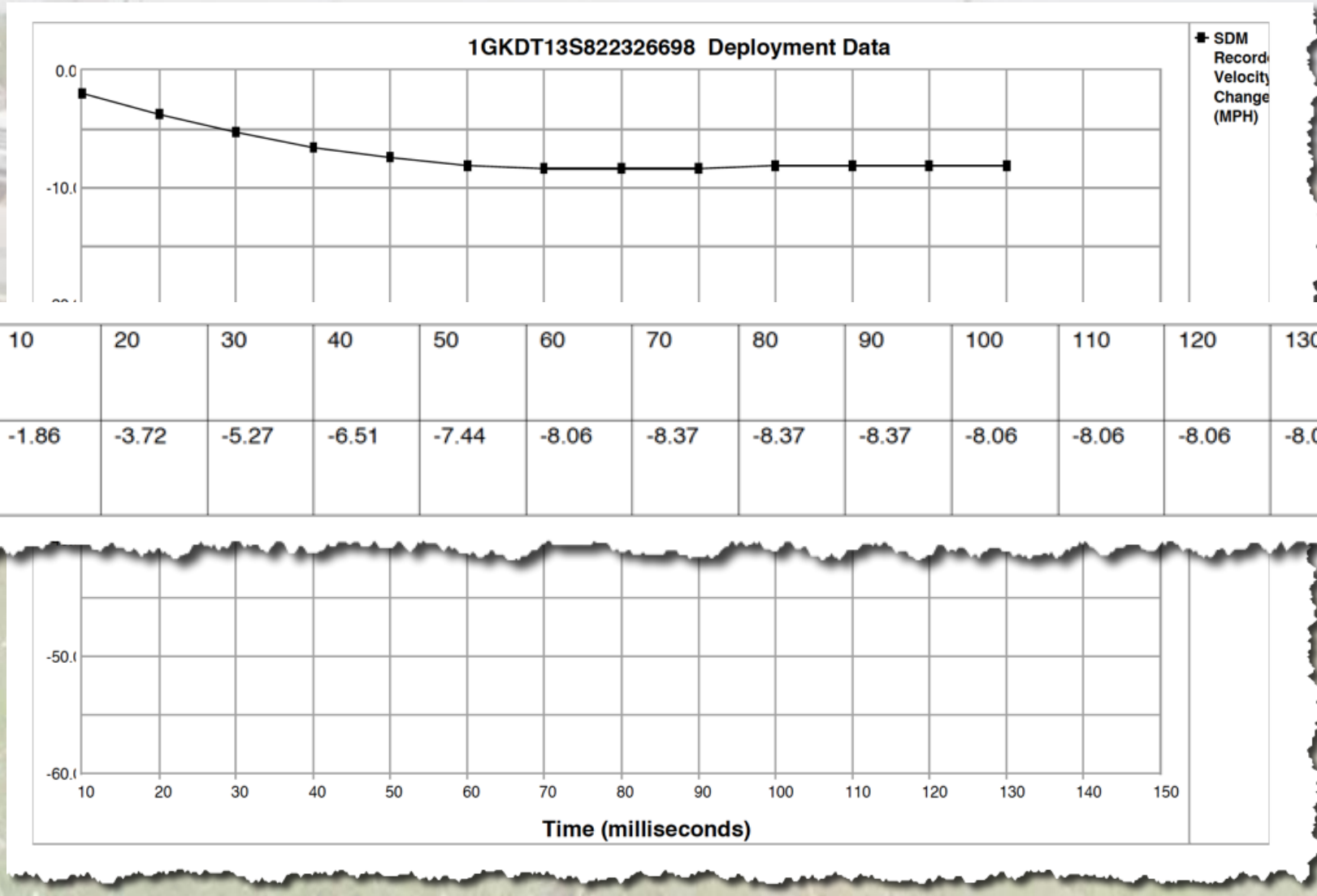
SIR Warning Lamp Status	OFF
Driver's Belt Switch Circuit Status	BUCKLED
Ignition Cycles At Deployment	26214
Ignition Cycles At Investigation	26215
Maximum SDM Recorded Velocity Change (MPH)	-8.46
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	85
Driver 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	32.5
Driver 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	35
Passenger 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	32.5
Passenger 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	35
Time Between Non-Deployment And Deployment Events (sec)	N/A
Frontal Deployment Level Event Counter	1
Event Recording Complete	Yes
Multiple Events Associated With This Record	No
One Or More Associated Events Not Recorded	No

CDR Data - Envoy

- The cable system means both vehicles had to be travelling at the same speed.
- The “-1” speed for the Rendezvous was 38 mph
- Highlights Data Timing with 1 second intervals

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle	Brake Switch Circuit State
				OFF
				OFF
				OFF
-5	26	1088	0	OFF
-4	30	960	0	OFF
-3	31	960	0	OFF
-2	34	960	0	OFF
-1	36	896	0	OFF

CDR Data - Envoy



Time (milliseconds)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
Recorded Velocity Change (MPH)	-1.86	-3.72	-5.27	-6.51	-7.44	-8.06	-8.37	-8.37	-8.37	-8.06	-8.06	-8.06	-8.06	N/A	N/A

Post Impact Speed Reliability?




STOP

What to work with?

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle	Brake Switch Circuit State
				OFF
				OFF
				OFF
-5	26	1088	0	OFF
-4	30	960	0	OFF
-3	31	960	0	OFF
-2	34	960	0	OFF
-1	36	896	0	OFF

No evidence of braking, steady acceleration.

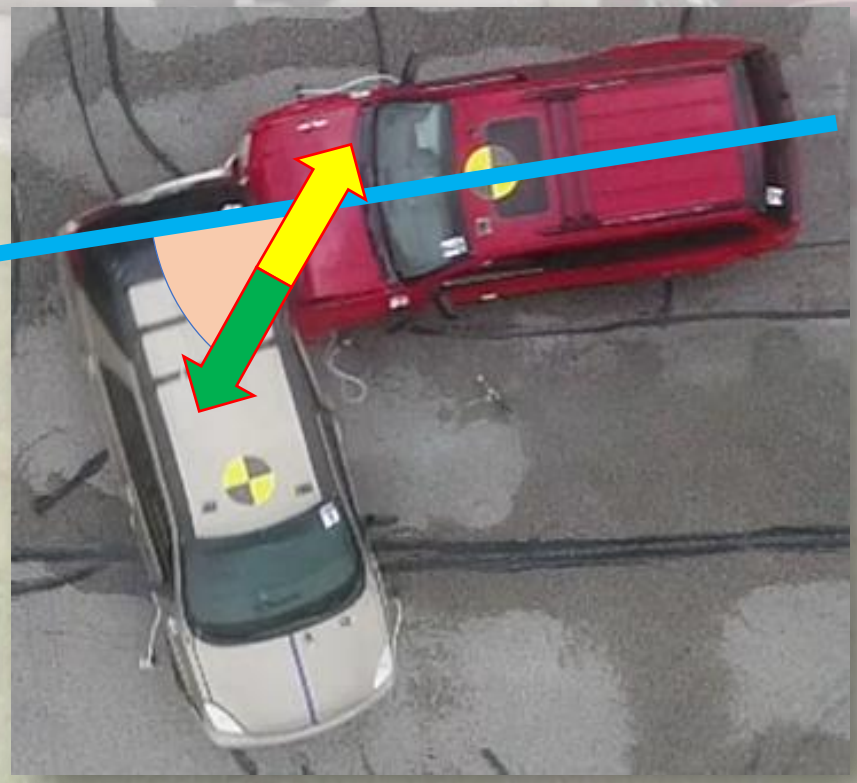
Even is 36 mph is a full second prior to impact,

There is no reasonable expectation it would vary much

Envoy Change in Velocity (x-axis)

Maximum SDM Recorded Velocity Change (MPH) -8.46

X-axis Change in Velocity has to be Adjusted for pdof to represent Δv along the pdof



$$\Delta v = \frac{\Delta v_x}{cos(pdof)}$$

$$\Delta v = \frac{8.46}{cos(45)}$$

$$\Delta v = 12\ mph\ (11.96)$$

Buick Change in Velocity

$$\Delta v_{Buick} = \frac{\Delta v_{Envoy} \times w_{Envoy}}{w_{Buick}}$$

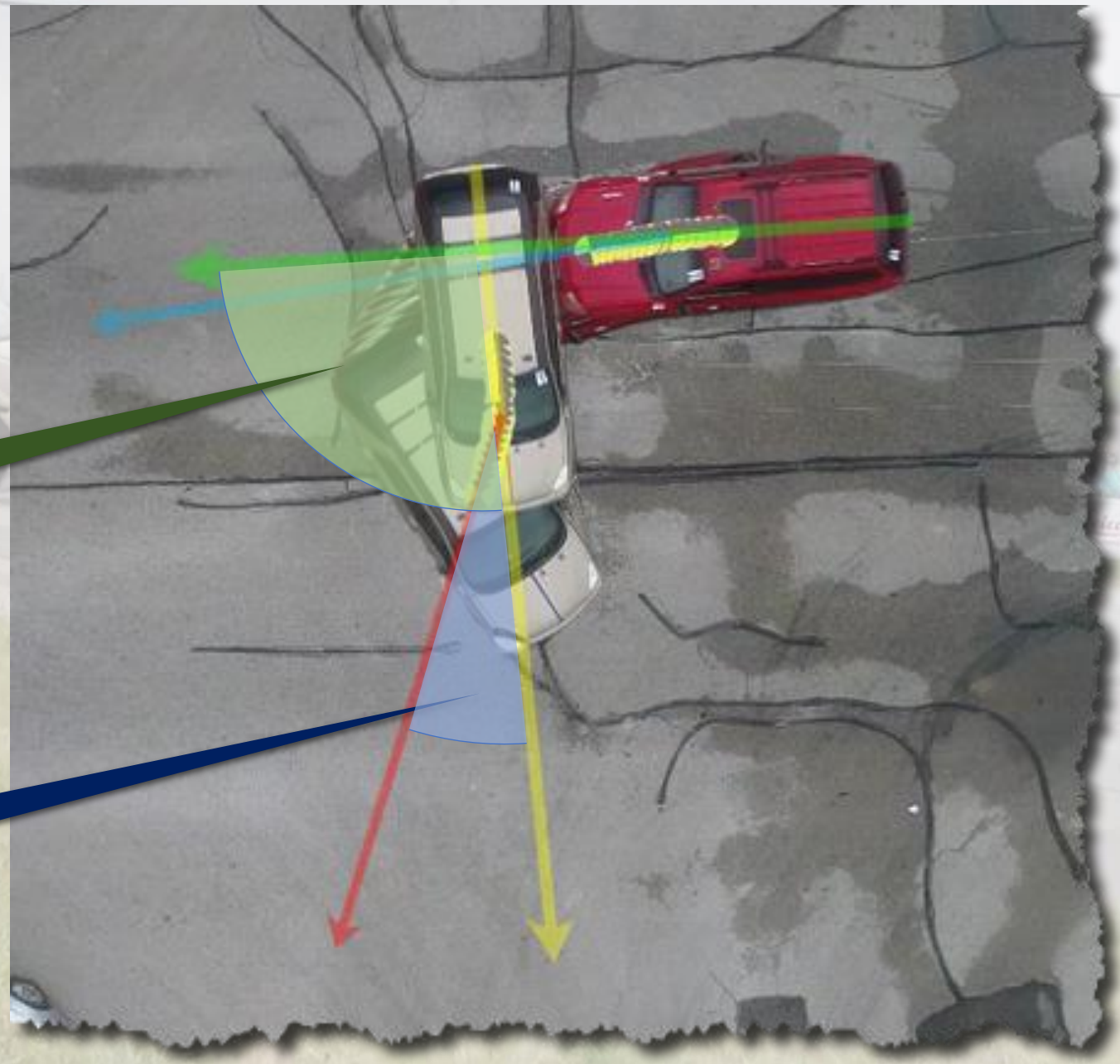
$$\Delta v_{Buick} = \frac{12 \times 4586}{3829}$$

$$\Delta v_{Buick} = 14.2\, mph$$

Angles?

Relative Approach Heading= 92˚

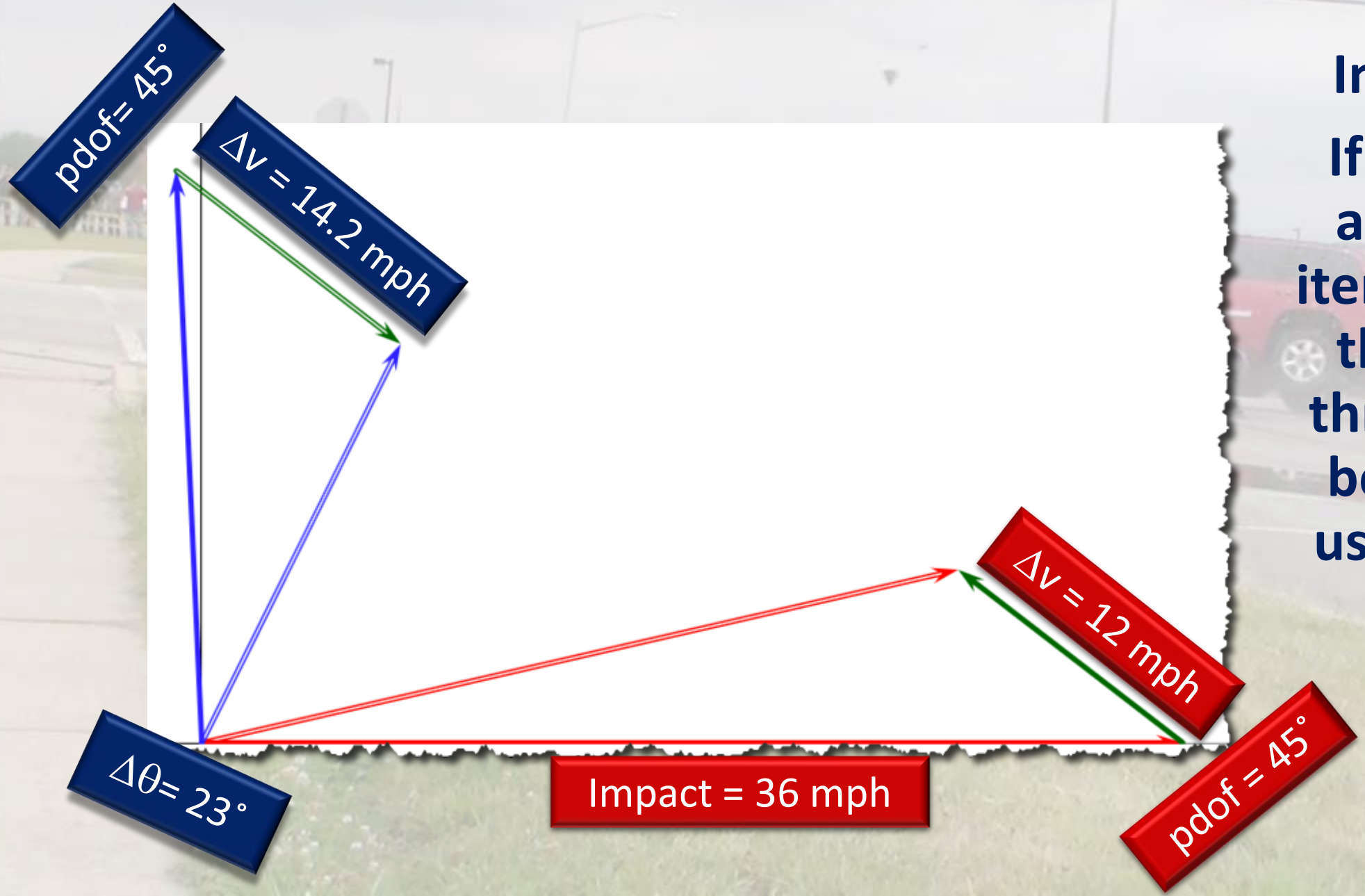
Buick Change in Direction = 23˚



pdof of the Buick

- The sum of the pdof values from each vehicle and their relative approach heading form a triangular relationship. Therefore the sum of these 3 angles must equal 180˚
- When the pdof of the Envoy (45˚) is added to the the relative approach heading (92˚) and then subtracted from 180˚,the result is the pdof of the Buick.

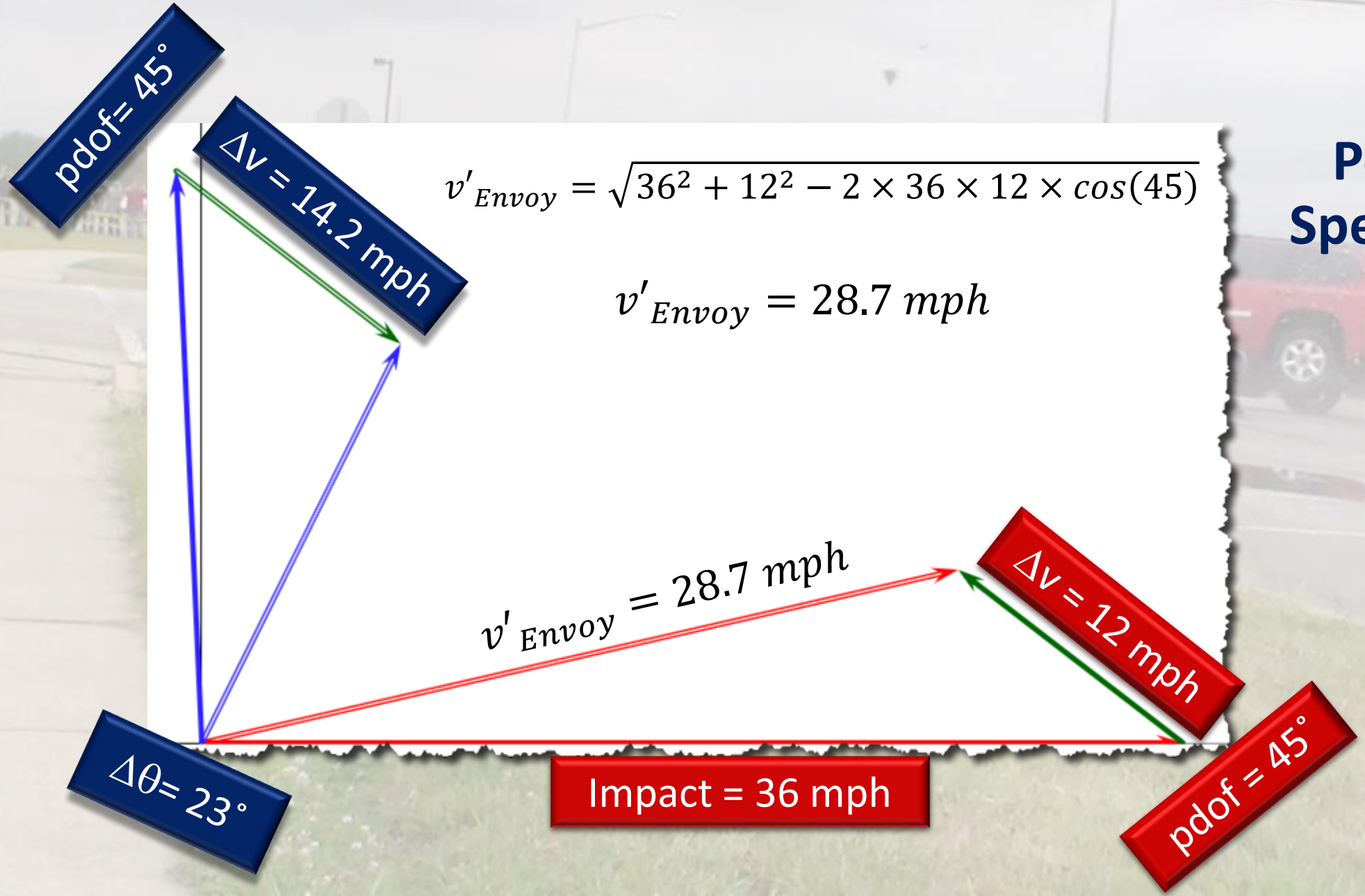
$$pdof_{Buick} = 180^{\circ} - (45^{\circ} + 92^{\circ}) = 43^{\circ}$$



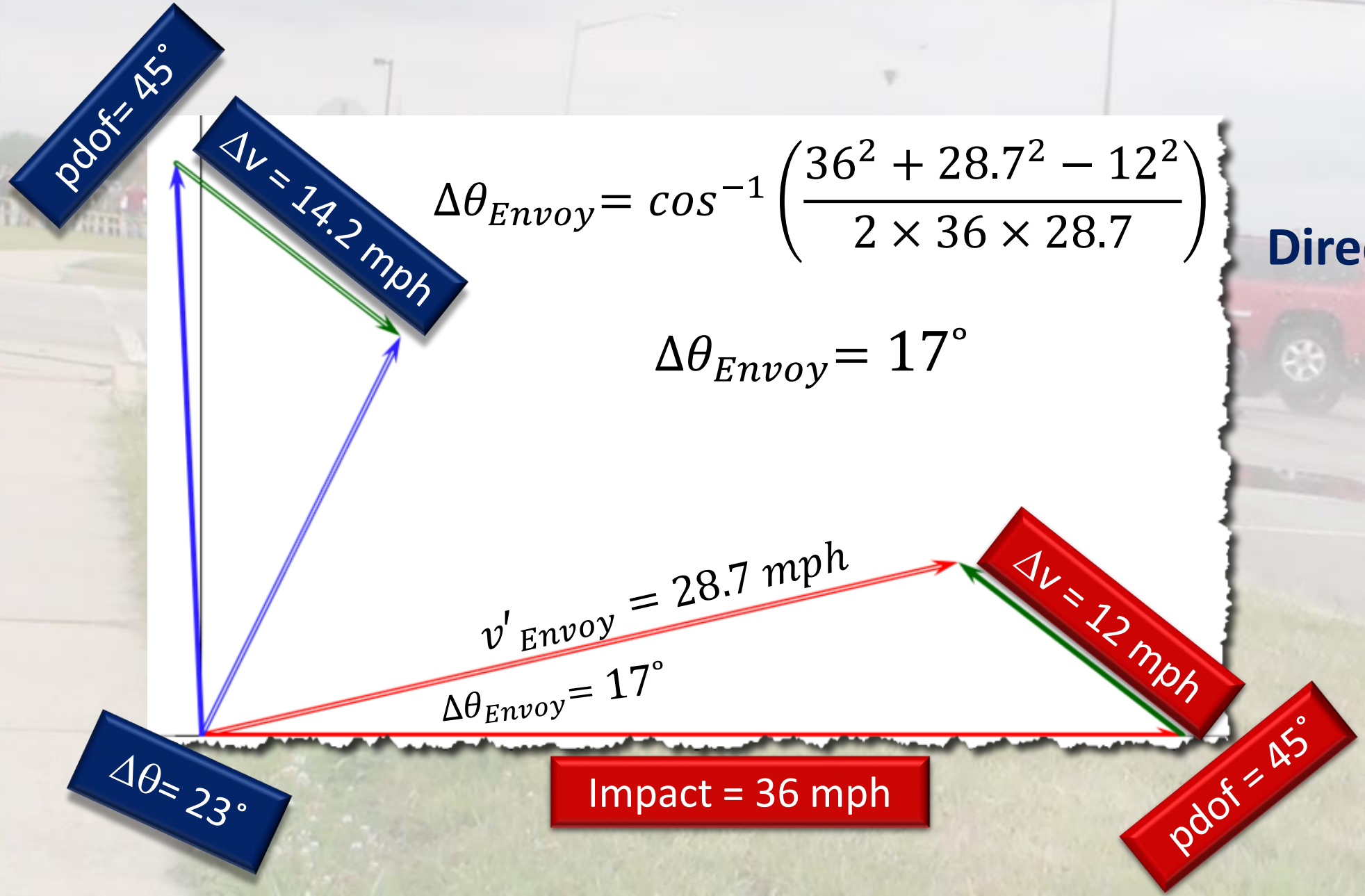
In any triangle

If one side and any two other items are known, the remaining three items may be determined using the law of sines and cosines.

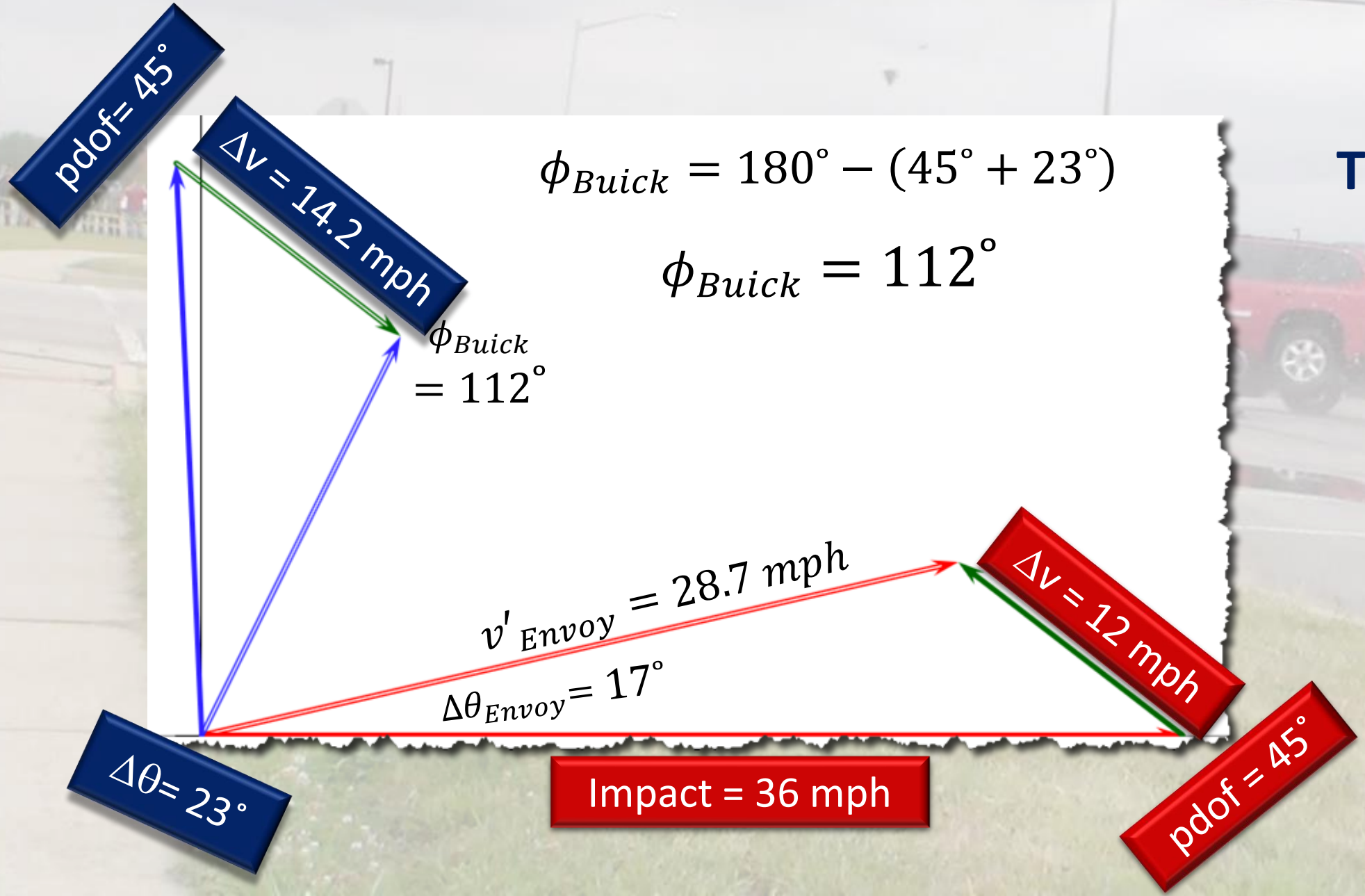
Post Impact Speed of Envoy



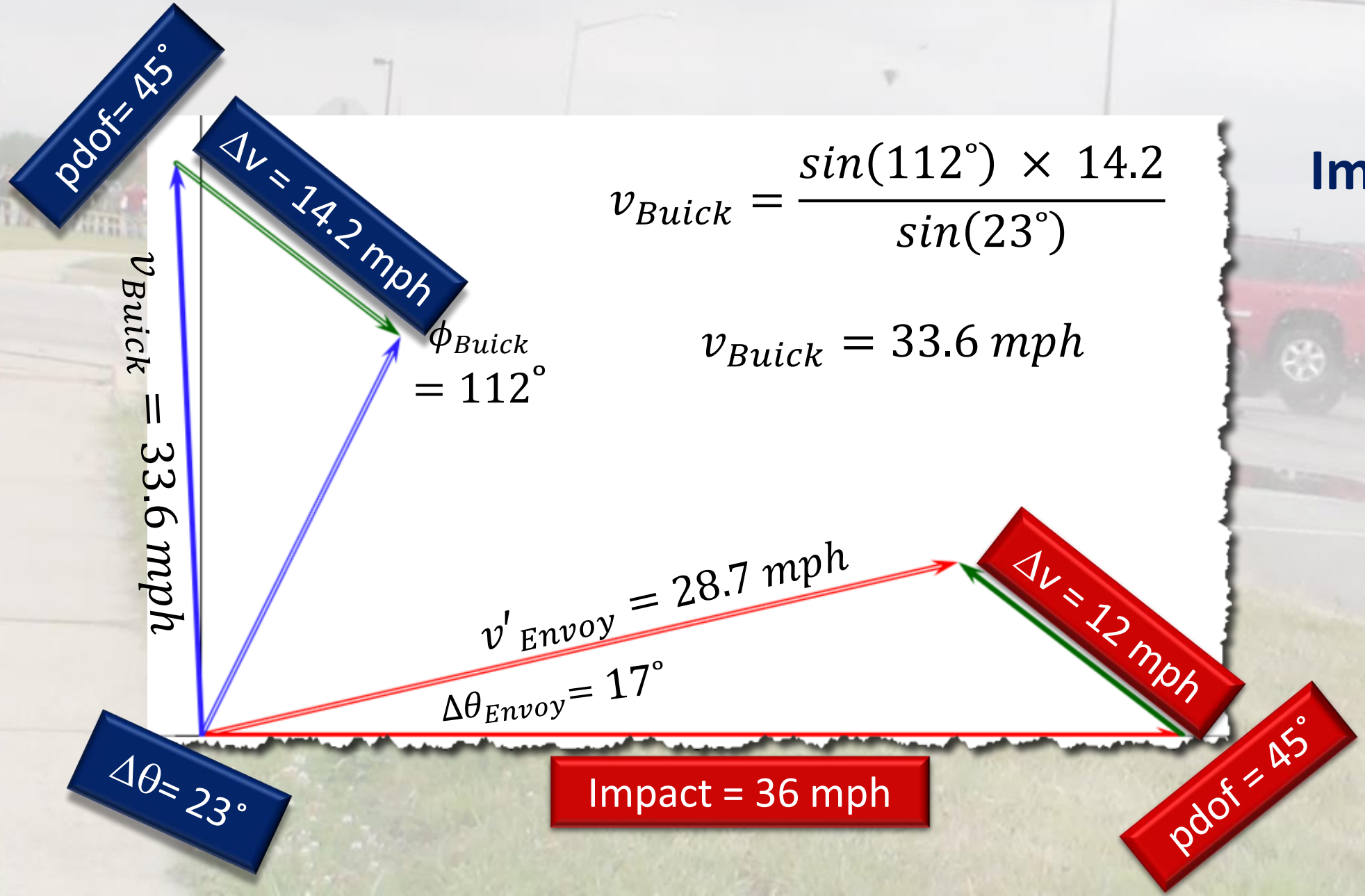
Change in Direction of Envoy



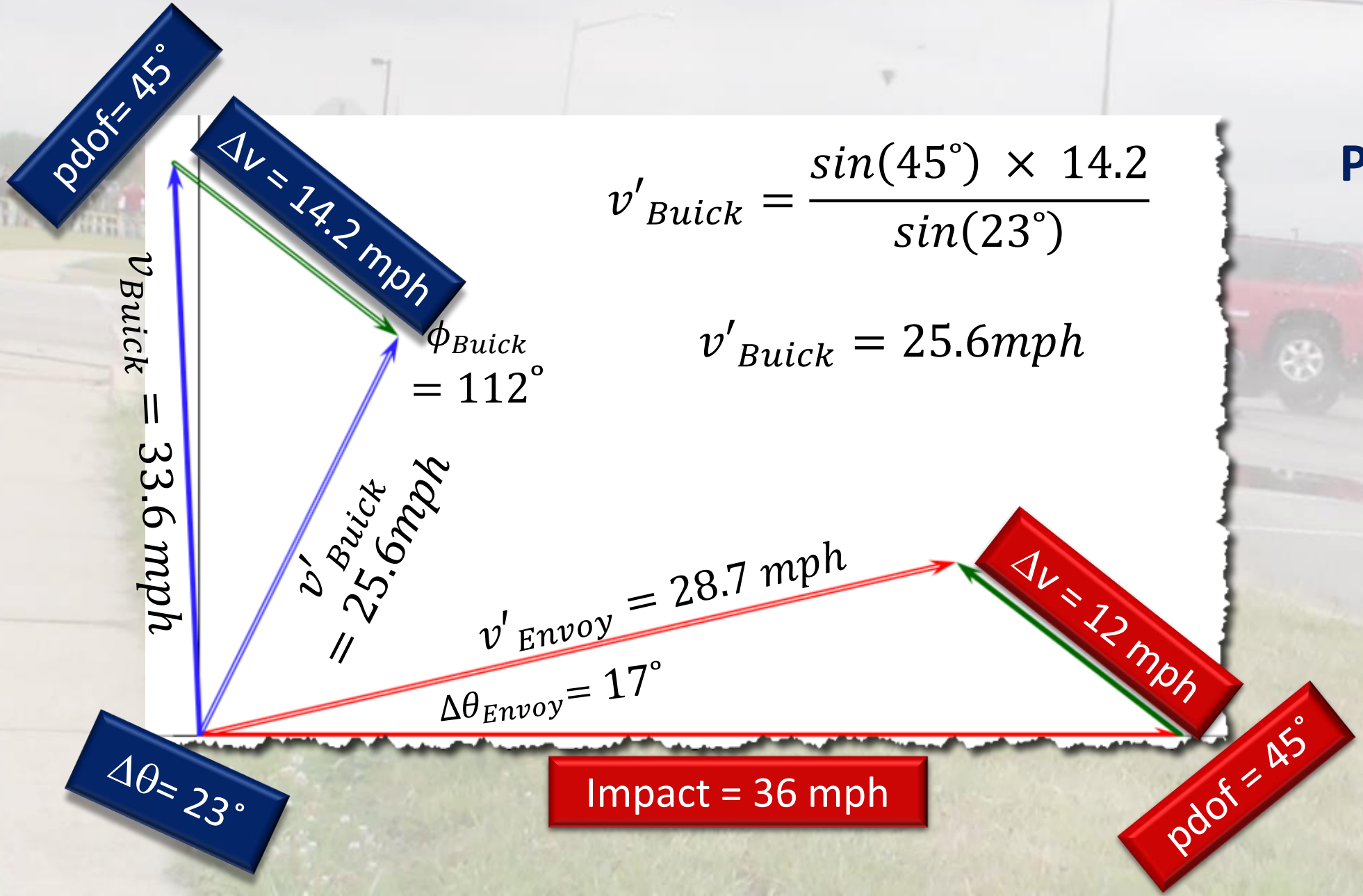
Third Angle Of the Buick



Impact Speed Of The Buick



Post Impact Speed of The Buick



$$v'_{Buick} = \frac{sin(45^{\circ}) \times 14.2}{sin(23^{\circ})}$$

$$v'_{Buick} = 25.6mph$$



IATAI 2017 – Crash Test #2

2002 Bonneville vs. 2006 Ford Freestyle vs. 2001 PT Cruiser

2002 Pontiac Bonneville

Actual Weight: 3834lbs – 4N6XPRT AutoStats Weight: 3700lbs



2002 Pontiac Bonneville

Actual Weight: 3834lbs – 4N6XPRT AutoStats Weight: 3700lbs



2006 Ford Freestyle

Actual Weight: 4065lbs – 4N6XPRT AutoStats Weight: 3959lbs





2001 Chrysler PT Cruiser

Actual Weight: 3051lbs – 4N6XPRT AutoStats Weight: 3112lbs

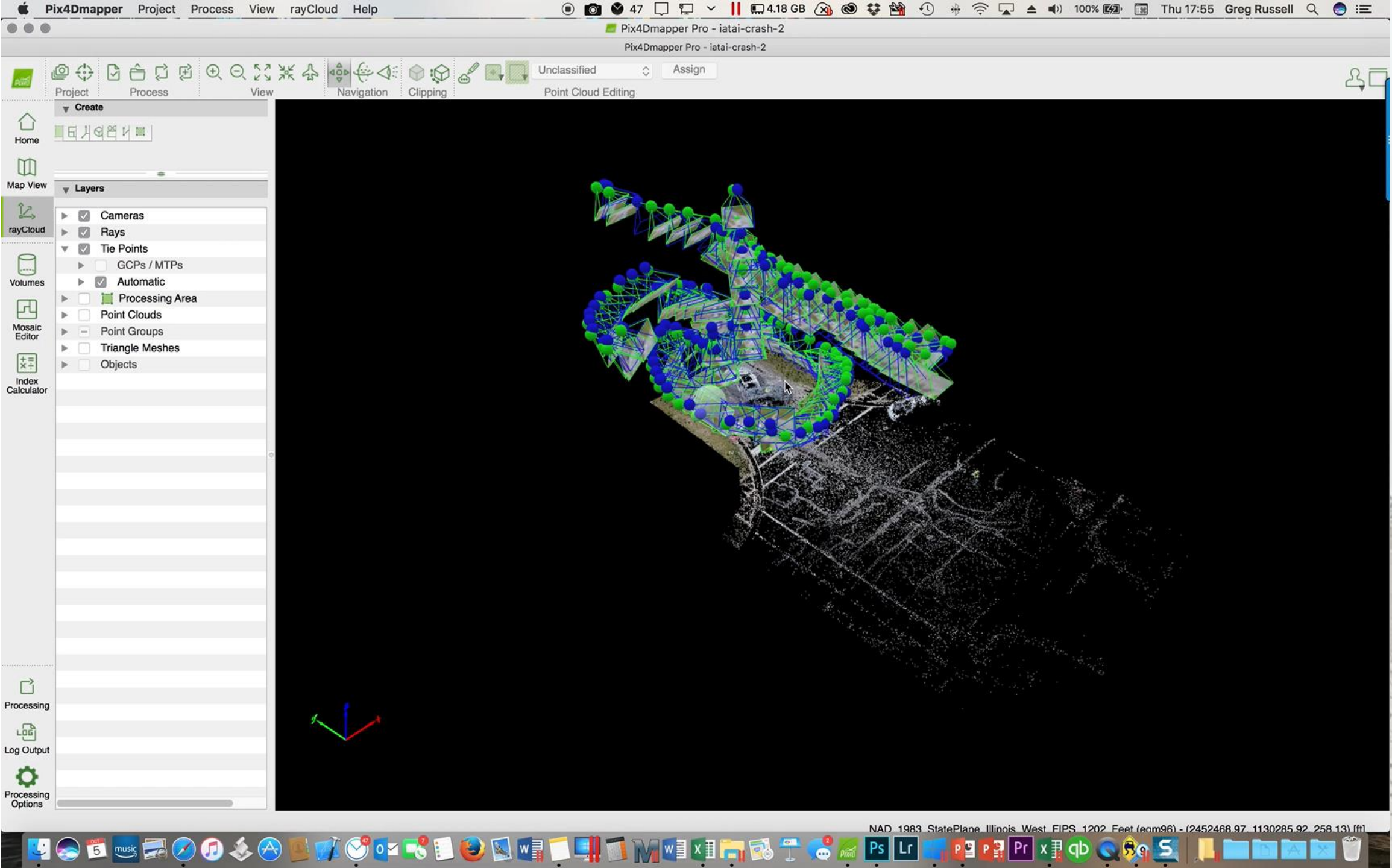


2001 Chrysler PT Cruiser



The Scene……




Pix4Dmapper Project Process View rayCloud Help
Thu 17:55 Greg Russell
Pix4Dmapper Pro - iatai-crash-2
Pix4Dmapper Pro - iatai-crash-2
Project
Process
View
Navigation
Clipping
Unclassified
Assign
Point Cloud Editing
Home
Map View
rayCloud
Volumes
Mosaic Editor
Index Calculator
Processing
Log Output
Processing Options
Create
Layers
Cameras
Rays
Tie Points
GCPs / MTPs
Automatic
Processing Area
Point Clouds
Point Groups
Triangle Meshes
Objects
NAD_1983_StatePlane_Illinois_West_FIPS_1202_Feet (egm96) - (2452468.97, 1130285.92, 258.13) [ft]


Residence
STOP

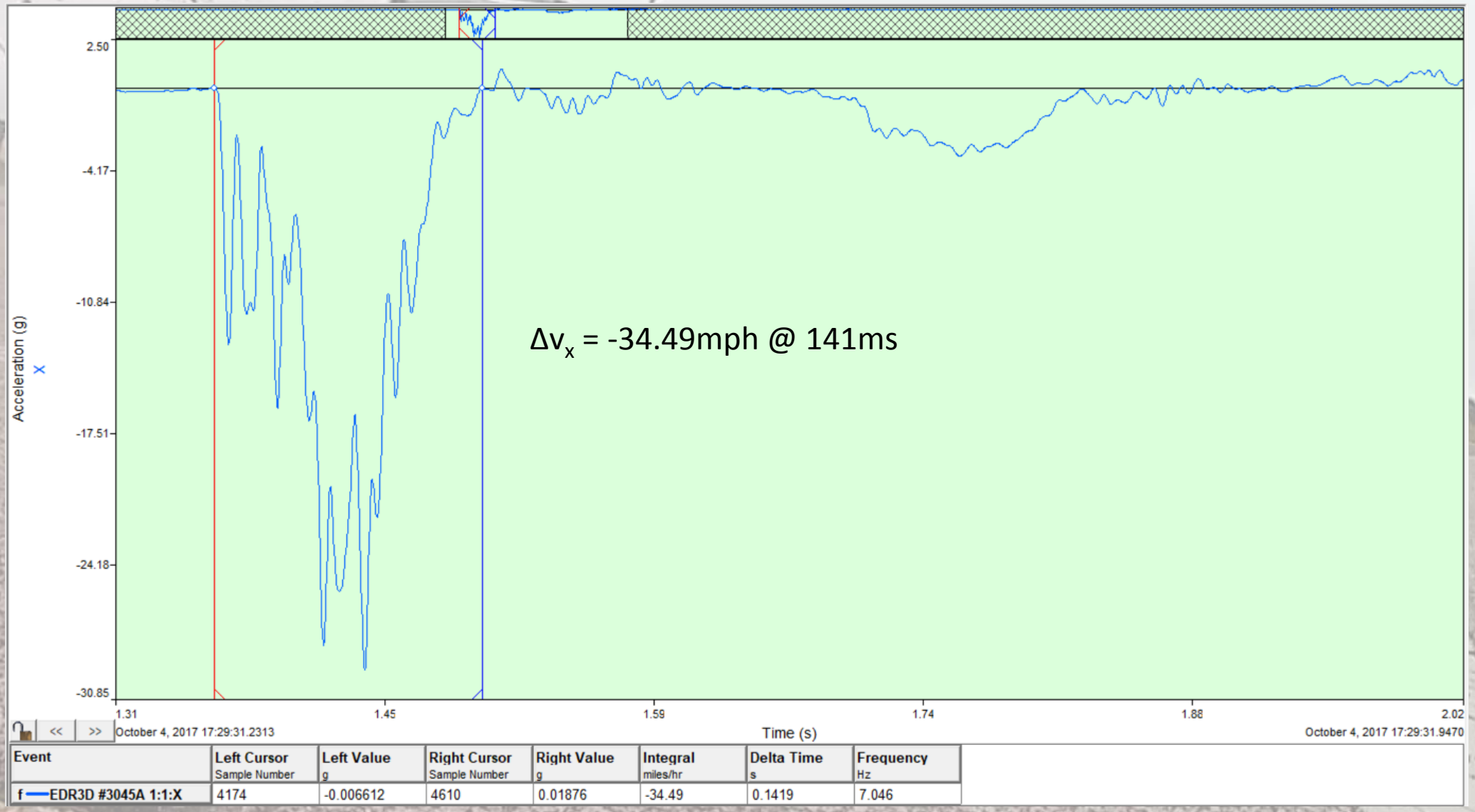






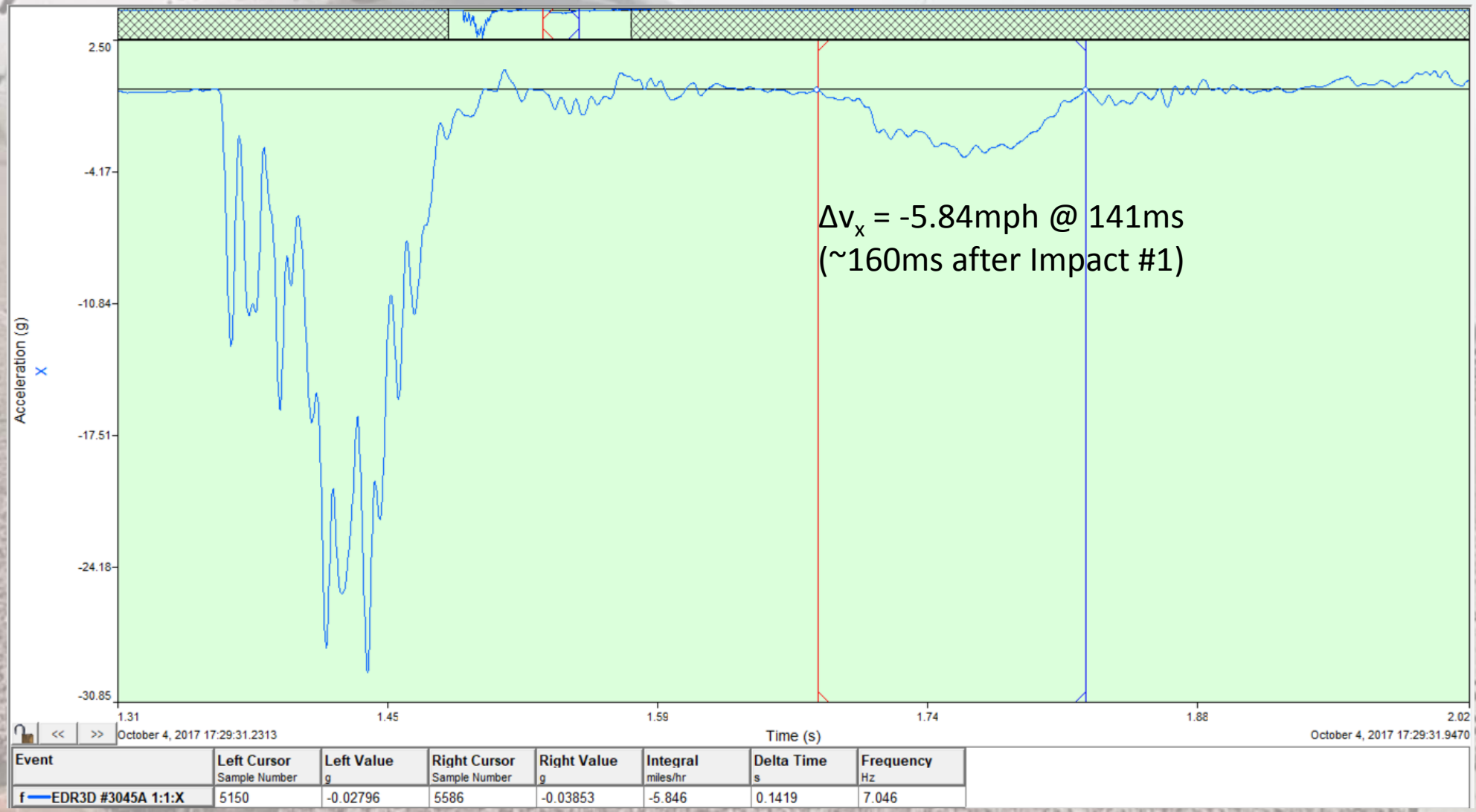

Leica

IST – Bonneville – Impact #1



Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f —EDR3D #3045A 1:1:X	4174	-0.006612	4610	0.01876	-34.49	0.1419	7.046

IST – Bonneville – Impact #2 - ~160ms after



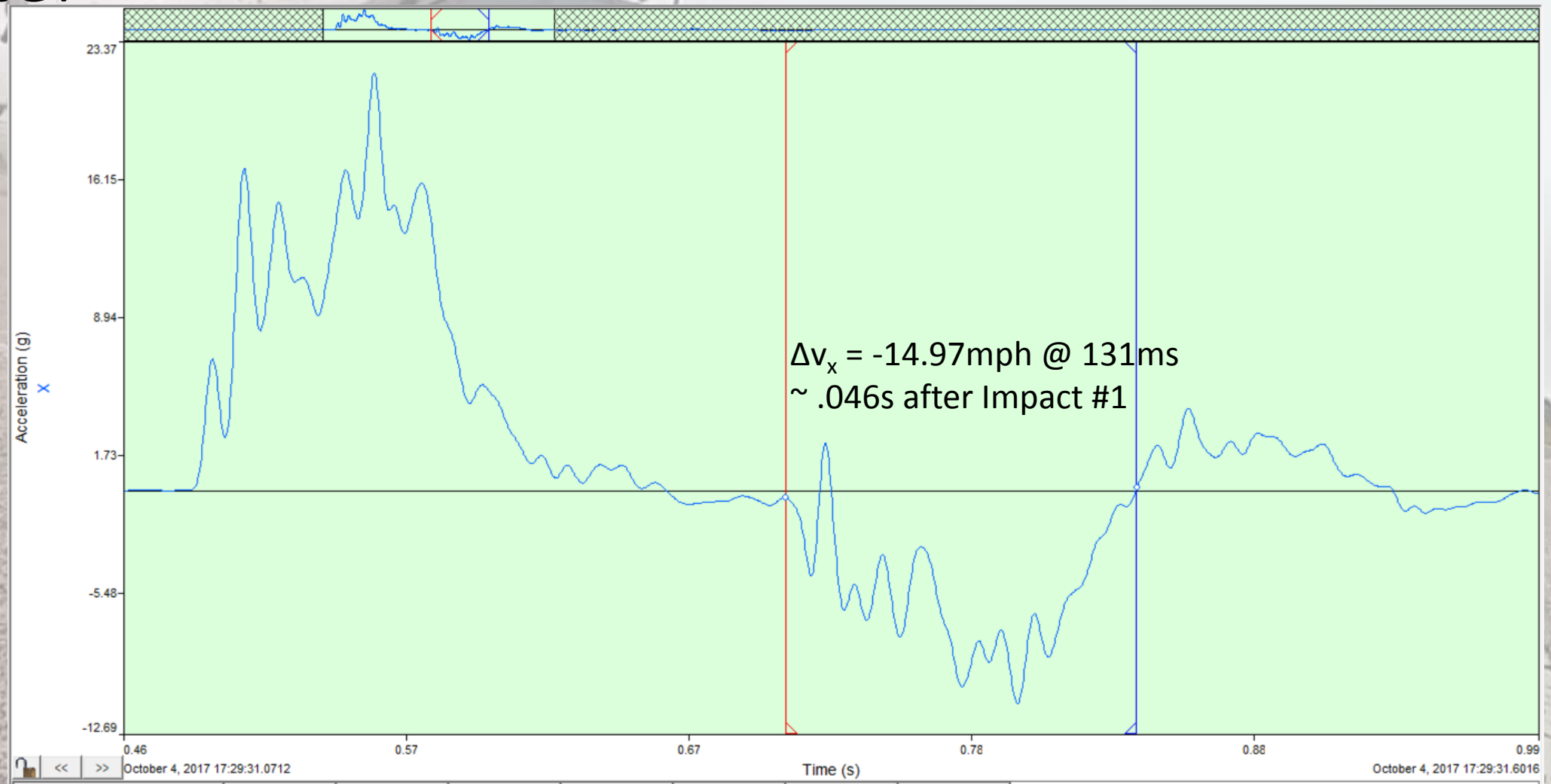
Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f —EDR3D #3045A 1:1:X	5150	-0.02796	5586	-0.03853	-5.846	0.1419	7.046

IST – Freestyle – Impact #1



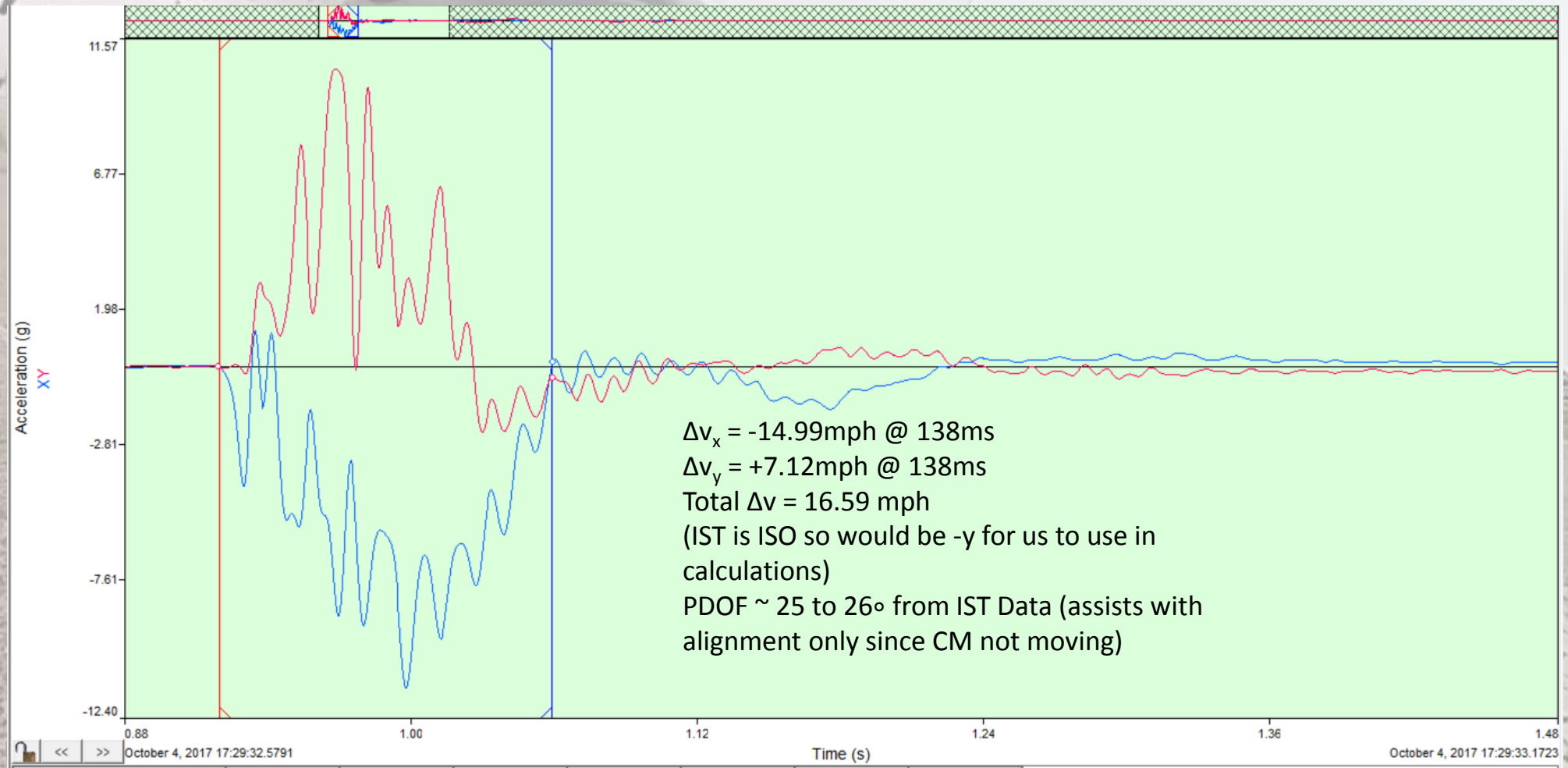
Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f —EDR3C #2211A 1:1:X	1477	0.04614	2036	0.05985	29.2	0.182	5.496

IST – Freestyle – Impact #2 ~.046 Seconds later



Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f —EDR3C #2211A 1:1:X	2175	-0.3773	2579	0.1466	-14.97	0.1315	7.604

IST – PT Cruiser



Event	Left Cursor Sample Number	Left Value g	Right Cursor Sample Number	Right Value g	Integral miles/hr	Delta Time s	Frequency Hz
f —EDR3C #2215A 1:1:X	2033	0.0136	2337	0.1437	-14.99	0.1382	7.235
f —EDR3C #2215A 1:1:Y	2033	-0.01747	2337	-0.3835	7.123	0.1382	7.235

CDR - Bonneville

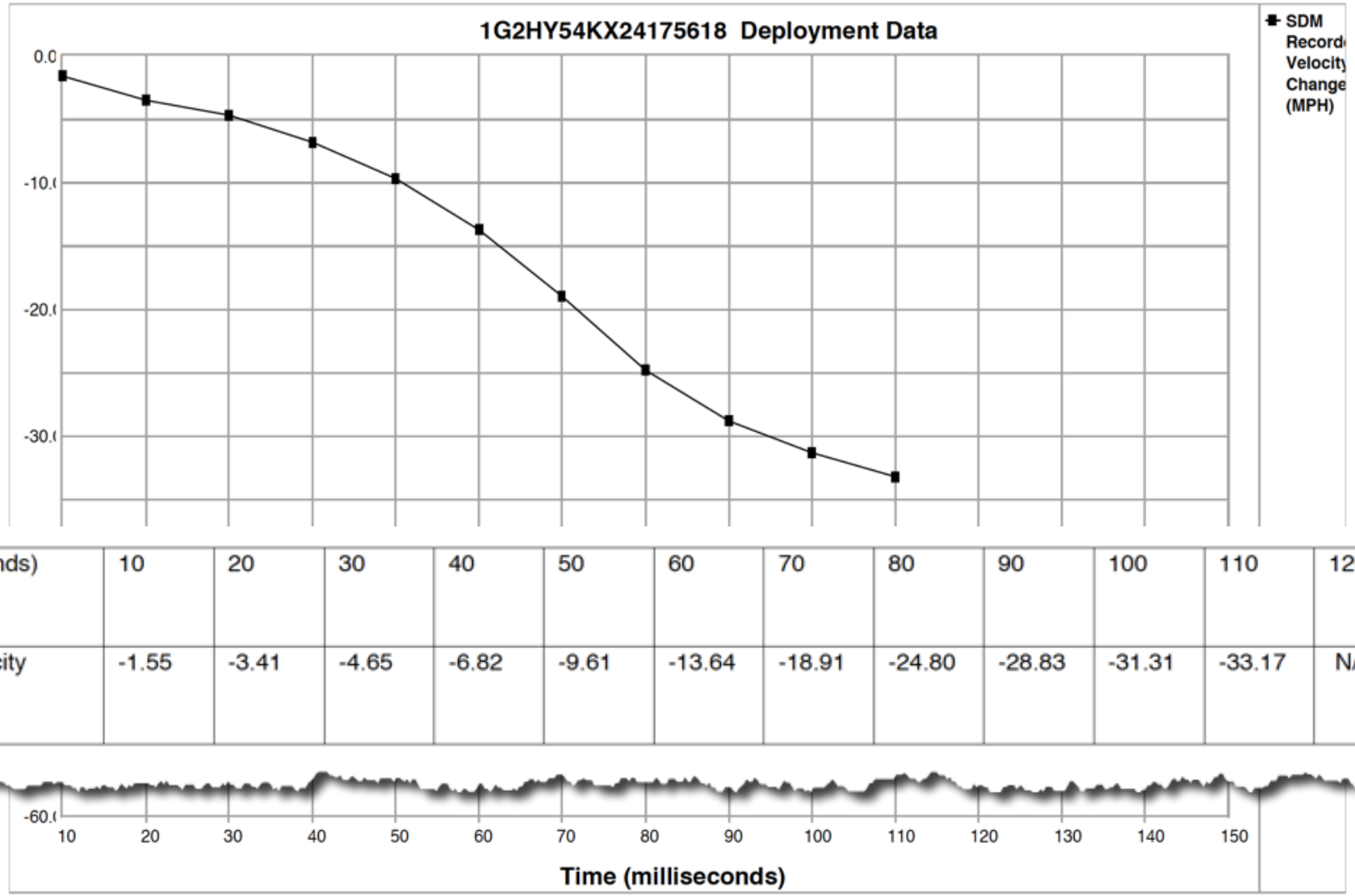
System Status At Deployment

SIR Warning Lamp Status	OFF
Driver's Belt Switch Circuit Status	BUCKLED
Ignition Cycles At Deployment	31814
Ignition Cycles At Investigation	31816
Maximum SDM Recorded Velocity Change (MPH)	-33.47
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	107.5
Driver 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	12.5
Driver 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	N/A
Passenger 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	12.5
Passenger 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	N/A
Time Between Non-Deployment And Deployment Events (sec)	N/A
Frontal Deployment Level Event Counter	1
Event Recording Complete	Yes
Multiple Events Associated With This Record	No
One Or More Associated Events Not Recorded	No

CDR - Bonneville

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle	Brake Switch Circuit State
				OFF
				OFF
				OFF
-5	44	832	0	OFF
-4	48	704	0	OFF
-3	52	960	0	OFF
-2	55	640	0	OFF
-1	57	832	0	OFF

CDR - Bonneville



Time (milliseconds)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
Recorded Velocity Change (MPH)	-1.55	-3.41	-4.65	-6.82	-9.61	-13.64	-18.91	-24.80	-28.83	-31.31	-33.17	N/A	N/A	N/A	N/A

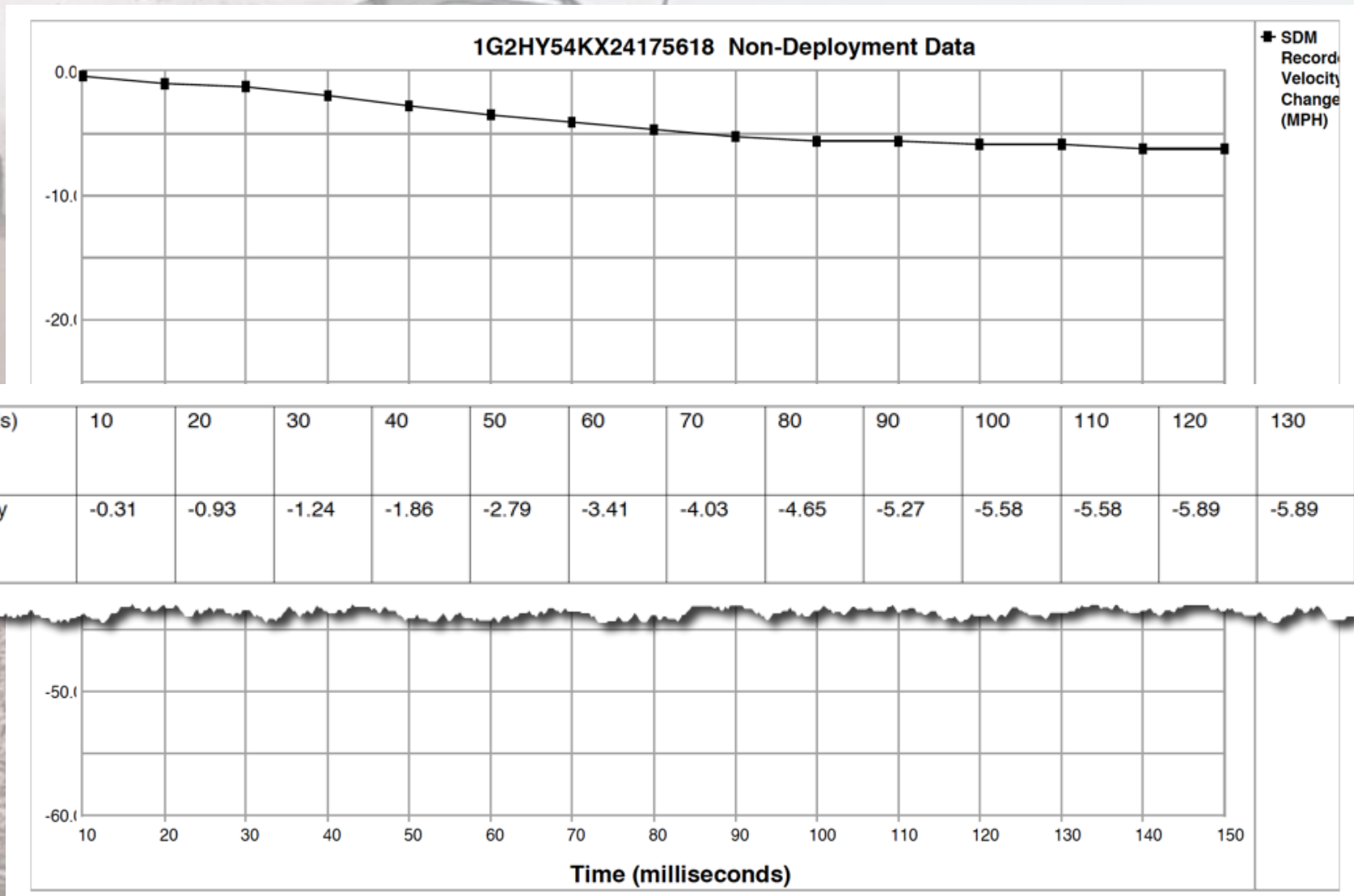
System Status At Non-Deployment

SIR Warning Lamp Status	ON
Driver's Belt Switch Circuit Status	BUCKLED
Ignition Cycles At Non-Deployment	31814
Ignition Cycles At Investigation	31816
Maximum SDM Recorded Velocity Change (MPH)	-6.75
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	172.5
Crash Record Locked	No
Event Recording Complete	Yes
Multiple Events Associated With This Record	No
One Or More Associated Events Not Recorded	No

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle
-5	48	704	0
-4	52	960	0
-3	55	640	0
-2	57	832	0
-1	59	768	0

Brakes = On ??!!!





Time (milliseconds)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
Recorded Velocity Change (MPH)	-0.31	-0.93	-1.24	-1.86	-2.79	-3.41	-4.03	-4.65	-5.27	-5.58	-5.58	-5.89	-5.89	-6.20	-6.20

Bonneville vs. Freestyle Δv Check

W_1: 3834lbs (Bonneville)
W_2: 4065lbs (Freestyle)
e: ?
Δv Bonneville: 33.49mph / 34.49mph

$$v_c = \frac{\Delta v_1 + \Delta v_2}{e + 1}$$

$$\Delta v_2 \times w_2 = \Delta v_1 \times w_1$$

$$\Delta v_2 = \frac{33.49 \times 3834 \ lbs}{4065 \ lbs}$$

$$\Delta v_2 = \frac{\Delta v_1 \times w_1}{w_2}$$

$$\Delta v_2 = 31.58 \ mph$$

$$\Delta v_{2\,IST} = 29.2 \ mph$$

Bonneville vs. Freestyle Calc'd Δv

W_1: 3834lbs (Bonneville)
W_2: 4065lbs (Freestyle)
e: ?
Δv Bonneville: 33.49mph / 34.49mph

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{34\ .49\ + 31\ .58}{1 + .10}$$

$$v_c = 60\ .06\ mph$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{34\ .49\ + 31\ .58}{1 + .05}$$

$$v_c = 62\ .92\ mph$$

Bonneville vs. Freestyle – IST Δv

W_1: 3834lbs (Bonneville)
W_2: 4065lbs (Freestyle)
e: ?
Δv Bonneville: 33.49mph / 34.49mph

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{34.49 + 29.2}{1 + .10}$$

$$v_c = 57.9\,mph$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{34.49 + 29.2}{1 + .05}$$

$$v_c = 60.65\,mph$$

Freestyle vs. PT Cruiser Δv Check

W_1: 4065lbs (Freestyle)
W_2: 3112lbs (PT Cruiser)
e: ?
Δv Freestyle: 14.97mph
V_1 Freestyle: 29.2 -31.58mph

$$v_c = \frac{Dv_1 + Dv_2}{e+1}$$

$$\Delta v_2 \times w_2 = \Delta v_1 \times w_1$$

$$\Delta v_2 = \frac{14\ .97 \times 4065\ \ lbs}{3112\ \ lbs}$$

$$\Delta v_2 = \frac{\Delta v_1 \times w_1}{w_2}$$

$$\Delta v_2 = 19\ .55\ mph$$

$$\Delta v_{2\ PT\ _\ IST} = 16\ .59\ mph$$

Freestyle vs. PT Cruiser – Calc'd Δv

W_1: 4065lbs (Freestyle)
W_2: 3112lbs (PT Cruiser)
e: ?
Δv Freestyle: 14.97mph
V_1 Freestyle: 29.2 -31.58mph

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{14.97 + 19.55}{1 + .10}$$

$$v_c = 32.74\ mph$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{14.97 + 19.55}{1 + .05}$$

$$v_c = 33.58\ mph$$

Freestyle vs. PT Cruiser IST Δv

W_1: 4065lbs (Freestyle)
W_2: 3112lbs (PT Cruiser)
e: ?
Δv Freestyle: 14.97mph
V_1 Freestyle: 29.2 -31.58mph

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{14.97 + 16.59}{1 + .10}$$

$$v_c = 30.05\ mph$$

$$v_c = \frac{\Delta v_1 + \Delta v_2}{1 + e}$$

$$v_c = \frac{14.97 + 16.59}{1 + .05}$$

$$v_c = 30.77\ mph$$

Conclusions:

- Starting with one vehicles Δv we can resolve for the impact speeds and Δv's of all of the vehicles
- Closing Speed from Δv is a valid method to calculate the impact speed(s) of the vehicles in a collision
  - Where one vehicle is stopped or the speed of one vehicle is known
  - Can be a valid double-check for the CDR reported vehicle speed
- CDR reported speeds even for 15 year-old vehicles continue to be an accurate reflection of that vehicles speed and Δv

